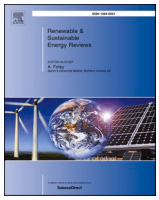
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A collated overview on the evaporative cooling applications for photovoltaic modules

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ABSTRACT

Cooling of photovoltaic modules is a crucial subject because it recovers and improves the module's electrical efficiency, power output, and lifetime, especially in hot and arid climates. Thus, different cooling methods have been innovated and investigated in many studies. Evaporative cooling is one of these applied methods, holding a great potential for managing the temperature of the modules especially installed in hot and dry lands. This study aims to inform researchers about current techniques which have been employed for evaporative cooling systems for PV modules. Moreover, it is the first review on evaporative cooling systems for photovoltaic modules. This review study highlights that existing evaporative systems mainly consist of techniques such as absorbent pads, synthetic clay, zeolite, phase change layers, co-current configurations, droplet evaporation, PV evaporative chimney, chimney-like channels, atmospheric water harvester and the heat pipe thermosiphon. As the result of the interpretations of the collected literature, it can be said that some of these applications have a great potential in enhancing an overall performance of a PV system. The current evaporative systems appear to be environmentally friendly, inexpensive, and reliable. They mainly utilize water as a coolant. Besides, it is recommended that different chemicals with low boiling points can be evaluated as alternative coolants to water. Moreover, many other recommendations for future studies are suggested.

1. Introduction

Solar irradiance makes it possible for life to exist in the planet earth. Apart from nuclear energy, it is the source of all energy supplies and one of the most adaptable sources of renewable energy applications [1–3]. Considering limited natural resources, increasing population and industrial growth, solar energy is one of the safest and promising solutions to energy and ecological problems presently being faced [4–12].

A photovoltaic (PV) system basically comprises of solar photovoltaic panels, storage units and inverters/chargers, as shown in Fig. 1 [13].

Photovoltaics (PVs) consist of small components called solar cells that utilize solar energy by transforming solar irradiation into direct current electricity and heat energy [9–19]. Monocrystalline silicon, Multicrystalline silicon, Advanced silicon designs, and Concentrator are commonly used solar photovoltaic cells [13]. Despite being the oldest photovoltaic technology, the most dominant photovoltaic technology is still the siliceous based photovoltaic technology having an average conversion efficiency between 10% and 15% [20]. In a PV panel, several cells are electrically attached in series and these series connections can be attached in parallel [21]. Fig. 2 demonstrates a structure of a typical PV module.

The most important aspect of PVs is their efficiency since they are associated with the amount of energy generated regarding the area occupied by the installation [22]. There are a lot of environmental factors such as temperature, irradiance intensity, wind velocity, dust, and humidity, that can affect the efficiency of a PV panel [17,23–25]. Jafari [19] indicates that a panels' efficiency ranges from 6% to 25% varying with the manufacturing methods and semiconductor materials. Elminshawy et al. [3] declares that 7%–40% of solar radiation can be converted into direct electricity according to photovoltaic cell production materials. Sargunanathan et al. [26] states that the commercially available PV panel exhibits conversion rates varying between 12% and 18%, while the laboratory solar cells achieve efficiency of 24.7%. Ghadikolaei [13] states that photovoltaic cells directly transform around 9–12% of incident sunlight to electrical energy, and over 80% of it is

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Nomenclature abbreviations

PV	Photovoltaic
PVs	Photovoltaics
CPV	Concentration Photovoltaic
STC	Standard Test Conditions
FTCC	Floating Tracking Concentrating Cooling System
PV/TE	Solar Photovoltaic/Thermoelectric
PV/T	Solar Photovoltaic/Thermal
LCAA	Life Cycle Assessment Analysis
$(C_2H_5)_2O$	Diethyl ether
(CS_2)	Carbon disulfide
$((CH_3)_2CO)$	Acetone
Units	
°C	Celsius degree
W	Watt
V	Volt
mm	Millimeter
cm	Centimeter
PCM	Phase Change Material
EC	Evaporative Cooling
RAFs	Rectangular Aluminum Fins
PAM	Polyacrylamide
CNT	Carbon Nanotube
$CaCl_2$	Calcium Chloride
PV/EC	Photovoltaic Evaporative Cooling
NTU	Number of Transfer Units
(CH_2Cl_2)	Dichloromethane
(C_2H_6O)	Ethanol
$(CH_3)_2CHOH)$	Isopropyl alcohol
W/m^2	Watts per square meter
L	Liter
ml	Milliliter
m/s	Meter per second

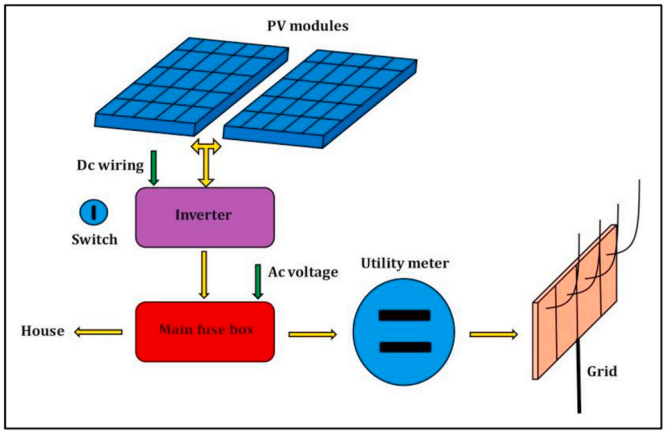


Fig. 1. A simple PV system [13].

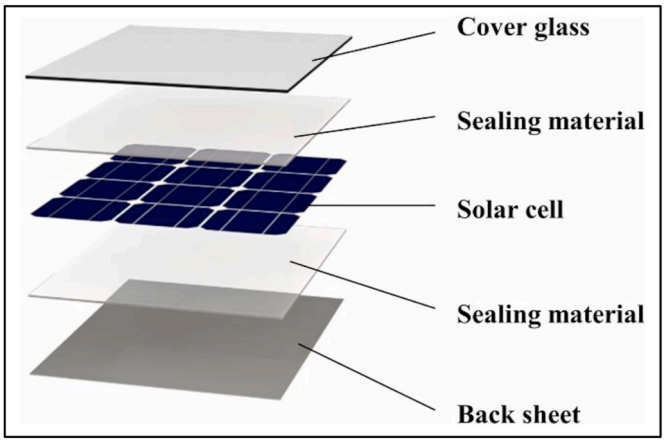


Fig. 2. Usual PV module structure [4].

wasted. In addition [14,24,27–33], claimed that approximately 10–20% of solar radiation can be directly harnessed and transformed into electricity by PV systems. Thus, a photovoltaic cell transforms a small portion of irradiance into electricity.

A novel technology of photovoltaic systems called Concentration Photovoltaic (also known as Concentrated Photovoltaic) (CPV) has been adopted in solar production. It increases the PV systems' efficiency and thus lowers the cost of energy generation [13,34]. CPV and solar tracking techniques improve the efficiency of a PV module by increasing the solar irradiance the module receives [22,35]. For instance, utilizing solar concentrators/lenses, and solar tracking could lower the cost of the photovoltaic system by 19% and 48%, respectively [1]. Fig. 3 displays laboratory efficiency of various photovoltaic technologies and their efficiency improvement over the years [36].

Even though some modifications can increase the solar irradiance on PVs, the solar energy not transformed into electricity is mostly heat energy, leading to heating of the photovoltaic cells. Consequently, the cells function at temperatures higher than ambient temperature [3,26]. Overheating occurs because photovoltaic cells do not have an ability to convert all irradiance spectral wavelength. Therefore, the spectral wavelength not transformed into electricity overheats the cells [3,12,37–39]. Since the conditions that PVs operate in are vital to their performance and lifetime due to high ambient temperature and constantly produced heat in the panels, the conversion efficiency and durability of the panels are reduced [3,38]. For instance, the conversion energy decreases around 4% when the temperature of the PV panel rises a Celsius degree (°C) over 25 °C [19]. Zhang et al. [6] stated that when the cell temperature surpasses 25 °C, there is a 0.26–0.50% decrease in the output power of the PV module. Also, it is reported that every 1 °C increase in PV cells temperature leads to approximately a 0.2–0.5% decline in the electrical power output of crystalline silicon PV cells [40–42]. Fig. 4 demonstrates a correlation between power output (in Watts) of a photovoltaic cell and voltage (in Volts) at different temperature [43].

A PV located in a hot environment with greater irradiation degrades faster, and consequently decreases in electricity generation. Although there is a negligible increase in the current of the cells when the temperature of the cells rises, voltage of the cells has a significant drop [44]. It is claimed that monocrystalline and polycrystalline photovoltaic modules performed 19.8% and 18.7% better, respectively, in the winter in comparison to the summer season [19]. For example, Chander et al. [45] carried out a study examining the impact of cell temperature on the PV characteristics of mono-crystalline silicon cells. The results indicated that higher cell temperature led to reductions in the open circuit voltage, maximum power, fill factor, and overall efficiency. Fig. 5 reveals a relation between electrical efficiency and cell temperature for a usual summer day [7,31,46].

Allegedly, rooftop solar panel related fires have recently been elevated due to the overheating of the hot spot that happens because of

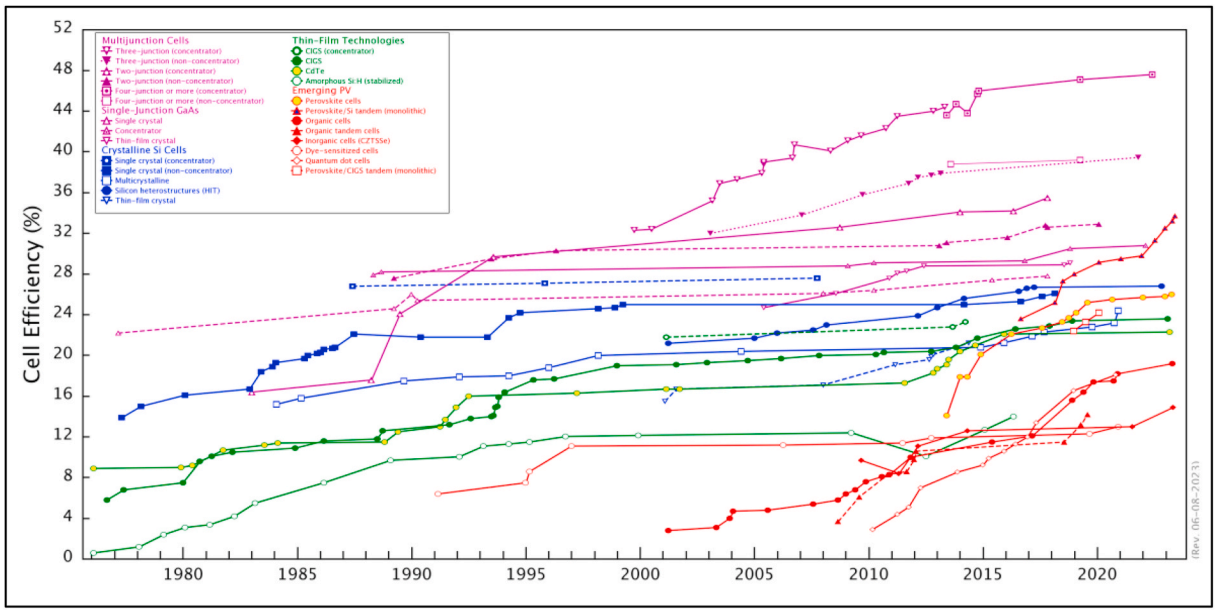


Fig. 3. Laboratory efficiency of various photovoltaic technologies and their efficiency improvement over the years [36].

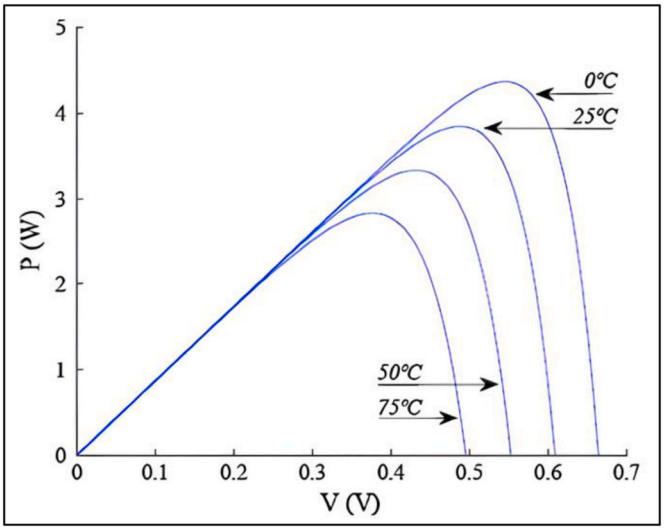


Fig. 4. Photovoltaic cell's power output at different temperatures [43].

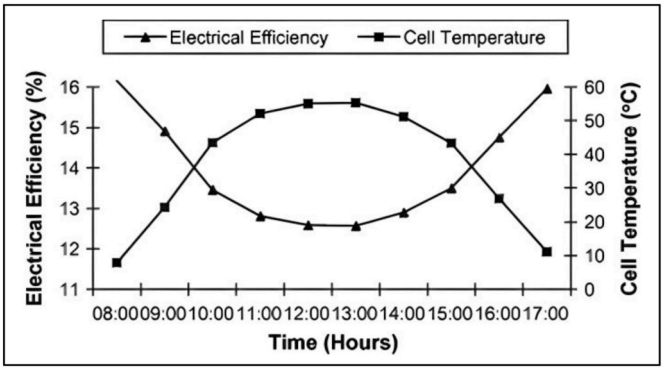


Fig. 5. Electrical efficiency rate and cell temperature's behavior on a summer day [7,31,46].

cracked cells, solar band failure, shading, and so on. The hot spot formation increases the temperature of a PV panel due to heat production rather than electricity generation and stimulates the degradation of the panel. Therefore, particularly in hot and arid climates, cooling is crucial to recover and improve PV panels' electrical efficiency, and reliability [19]. The cooling increases the power output and performance of the panels. Moreover, cooling systems have a potential to decrease the cost of electricity generated from PVs because, in addition to an increase in the efficiency, and output, it makes possible to use solar concentration, and prolongs photovoltaic cells' lifetime. The heat extracted by the cooling systems can be utilized for domestic and industrial applications, lowering the cost of energy production [4,44]. However, how the cooling is done is critical because non-uniform cooling of a PV module can lead to series issues such as temperature variations through the cells [46]. Temperature variations affect the whole panel and limits efficiency since the cells are linked in series. This case is known as the current matching problem [21]. Bahaidarah et al. [46] indicates that non-uniform temperature over the surface of a PV escalates cells' temperature and series resistance and reduces fill factor and conversion efficiency. Moreover, it causes of a formation of hot spots on the cells, resulting in lasting structural damage [21,46]. Therefore, a cooling technique should be used to keep a cell temperature at its lowest with constant distribution.

In order to decrease the temperature of PV panels, different cooling methods are innovated and conducted [6,19,29,47–49]. It is essential to design or select the optimum cooling method based on the type of material utilized in manufacturing PV panels and the climate conditions of the area the panels operate at [3]. A cooling system should be simple, have low maintenance cost, contain minimum toxic materials because of health and ecological concerns, and be reliable [21]. Classification of the cooling techniques can be made according to their phase systems, the type of working fluid, including cogeneration or not, active or passive cooling, and having moving components or no moving components. Also, a cooling system for a photovoltaic module can be designed based on cell temperature, constancy of temperature, maintenance expense, reliability, power required to circulate the cooling fluid, and capital expense [21,44]. The cooling systems are generally assigned into two categories: active cooling, and passive cooling [17,50–52].

A review manuscript published by Siecker et al. [47] provided a comprehensive overview of various PV cooling approaches designed to alleviate the adverse consequences of heightened temperatures, while

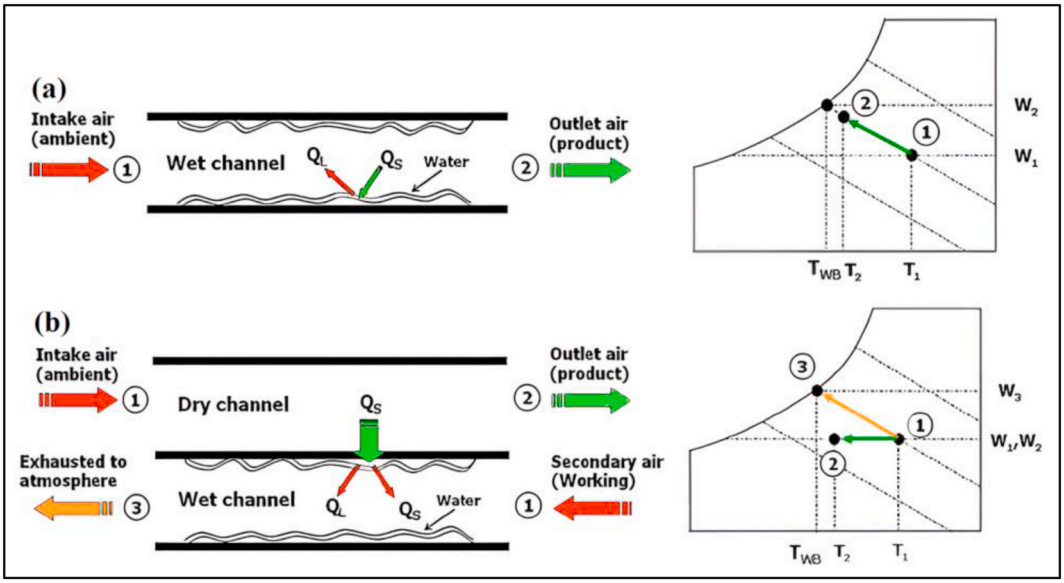


Fig. 6. (a) Direct EC; (b) indirect EC [53].

endeavoring to optimize the efficiency of PV solar panels functioning beyond the conditions recommended by Standard Test Conditions (STC). A range of cooling technologies was examined, including the Floating Tracking Concentrating Cooling System (FTCC); the Hybrid Solar Photovoltaic/Thermal System utilizing water spraying for cooling; the Hybrid Solar Photovoltaic/Thermoelectric (PV/TE) system incorporating heat sinks for temperature control; the Hybrid Solar Photovoltaic/Thermal (PV/T) with forced water circulation for cooling; strategies for enhancing solar panel performance through the incorporation of phase-change materials; the Water Immersion Cooling Technique for solar panels; the application of transparent coatings (photonic crystal cooling) for cooling Solar PV panels; the Hybrid Solar Photovoltaic/Thermal System cooled by forced air circulation, and the utilization of Thermoelectric cooling in solar panels. This review aimed to offer a comprehensive understanding of each of the technologies employed to mitigate the surface temperature of PV modules. However, the evaporative cooling applications have not been included in the review. In addition, a review study written by Salameh et al. [48] focused on the progress observed across various generations of existing solar PV technologies, placing particular emphasis on the management of solar panel temperature through a range of cooling methodologies. This study provided insights into the pairing of specific PV panels with corresponding cooling techniques, with a special focus on PV/T (photovoltaic/thermal) systems. The discussion extends to encompass a life cycle assessment analysis (LCAA) for both PV and PV/T systems, addressing environmental and economic considerations. Furthermore, Cuce and Riffat [53] presented another comprehensive review explored the various evaporative cooling technologies and their applications in the building systems. It was reported that evaporative cooling emerges as a groundbreaking technology, not only delivering cool air to occupants but also offering a promising avenue for curtailing carbon emissions and reducing energy consumption. The findings underscored the significant potential of evaporative cooling systems to yield substantial energy savings, particularly in hot and arid climates. However, the finding in this review study belongs to studies made at least a decade ago or far past. Moreover, an article by Hamzat et al. [54] provided a thorough literature survey concentrating on recent advancements in cooling technologies designed for solar PV systems. It extensively explored the influence of nanofluids on the efficiency of PV thermal systems, conducts analyses of prevailing studies on PV cooling techniques, investigates the application of nanofluids as both coolants and optical filters, and scrutinizes the synergistic effects of nanofluids and nano-phase change material (PCM) on the holistic performance of photovoltaic systems. This comprehensive review delivered a meticulous examination of various aspects within the domain of solar PV cooling, elucidating emerging trends and advancements in the field.

In present novel review study, we focused on the latest advancements in evaporative cooling studies for PV technologies, with a particular focus on the most innovative application of evaporative cooling methods. Apart from the comprehensive overview provided in existing literature [47,48,53,54], present study explores the potential and the most recent advancements of evaporative cooling aimed to further enhance the efficiency of PV solar panels. By introducing the novel elements of evaporative cooling applications, this study aims to contribute fresh insights into sustainable temperature control strategies for PV modules, addressing both experimental and numerical considerations. Moreover, it contributes to cumulative concept of the discipline and is the first review of EC systems for PV modules.

1.1. *Evaporative cooling (EC)*

Evaporation occurs when mass transfers from a liquid state to a vapor state. This process happens at temperatures below the boiling point. Similar to any other mass transfer process, the concentration difference is the driving force for evaporation. Mass generally moves from areas of high concentration to areas of low concentration. Additionally, by extracting the high concentration fluid near the surface and exchanging it with lower concentration fluid, the movement of fluid enhances mass transfer [55]. Relative humidity, air temperature, air movement, and expose surface area are the main factors affecting evaporation. They generally contribute to each other to regulate both the overall rate of evaporation and the cooling effect [56].

EC happens when a small portion of coolant changes phase and extracts unwanted heat away from the surface [57]. It refers to a process of latent heat cooling where water evaporates to create a state of balance with water vapor in the surrounding air, resulting in a cooling effect [55]. This cooling effect can be utilized to decrease PVs' temperature. EC is a well-known concept which has been used in various residential and industrial applications [55,58]. It is noncomplex, sustainable, affordable, effective, and eco-friendly cooling technique [30,59,60]. It is highly effective in hot and dry environments [53]. Therefore, it holds great potential for managing the temperature of PV panels installed in hot and arid climates [55]. Conversely, EC systems are not suitable for humid climates because the air is close to saturation, resulting in limited

Table 1
Experimental studies on EC of PV modules.

Study	System description	Systems category	Range of solar radiation (W/m^2)	Coolant	System performances
Suresh et al., 2013 [56]	Low-Cost EC System with Cooling Pad and Air Blow	Active	–	Water	Decrease in module's temperature A 10% increase in power output of solar PV system
Chandrasekar et al., [2013] [62]	EC System Using Cotton Wicks	Passive	–	Water	A 30% reduction in module temperature Increase in module efficiency
Alami, 2014 [57]	EC System Utilizing Synthetic Clay	Passive	900-905 ± 5	Water	Reduction in panels' temperature (reduction of 40 °C could be reached) An average increase of 15% in output voltage and an average enhancement of 16.7% in output power
Chandrasekar and Senthilkumar, 2015 [63]	EC System Using Heat Spreaders in Conjunction with Cotton Wicks	Passive	500-1000 ± 40	Water	A 12% reduction in PV module temperature A 14% increase in electrical power production
Lucas et al., 2017 [61]	PV Evaporative Chimney	Active	700 ± 50	Water	Temperature reduction by 8 °C 4.9–7.9% average increase in electrical efficiency.
Haidar et al., 2018 [55]	EC System Using Cloth and Rubber Pipes	Passive	400–1000	Water	Reduction of over 20 °C in PV panel temperature. Approximately 14% improvement in electrical power generation efficiency
Lucas et al., 2019 [64]	A Modified Version of the PV Evaporative Chimney	Active	400–1000	Water	An average temperature reduction of 15 °C A 10% enhancement in electrical efficiency
Hasan et al., 2019 [65]	EC with Wood Wool and Heat Sink	Passive	400–1000	Water	Around 27% drop in panel temperature A 32.7% increase in the panel's power output and a 31.5% improvement in module efficiency
Abdallah et al., 2019 [66]	EC System Using Zeolite	Passive	600–1000	Water	A temperature reduction ranging from 9 °C to 14.9 °C An average electrical improvement between 6.5% and 10%
Wang et al., 2019 [67]	Active Phase Change EC System Based on Porous Media	Active	500–900	Ethanol	A 19.32% increase in electrical power output A 18.40% improvement in PV module's efficiency
Ruiz et al., 2020 [68]	Evaporative Chimney to power a water-cooled chiller	Active	200–1000	Water	Increase in electrical efficiency of 12.62%.
Haidar et al., 2020 [69]	EC System Utilizing Co-Current Configuration	Active	300–1000	Water	A temperature reduction of over 10 °C A 5% enhancement in power output
Kadhim and Aljubury, 2020 [70]	EC Using Water from an Underground Tank	Active	600–1000	Water	Maximum 16% enhancement in power output and electrical efficiency Maximum average of 12.6 °C reduction in PV module's temperature
Li et al., 2020 [71]	EC System Based on Atmospheric Water Harvester	Passive	1000	Water	A minimum 10 °C decrease in PV panel temperature in laboratory conditions
Dida et al., 2021 [72]	Gravity-Assisted EC System Using Capillary Action of Burlap Cloth	Passive	200–800	Water	A 20 °C reduction in the PV module temperature (equivalent to 26% reduction) A 14.75% improvement in the electrical efficiency
Almuwailhi and Zeitoun, 2021 [73]	Evaporative natural cooling, and evaporative forced cooling utilizing a wetted fabric	Passive and Active	1000	Water	A 2.7% increase in efficiency for passive EC A 3.8% increase in efficiency for active EC
Chandan et al., 2022 [74]	EC System Using Cotton Wicks	Passive	300–1000	Water	10–22% drop in PV module temperature 11.3% increment in electrical efficiency
Malvika et al., 2022 [75]	Gravity-Assisted EC System Using Capillary Action of Burlap Cloth	Passive	650–935	Water	Up to 21.1 °C drop in the PV module temperature Up to 21% of power improvement
Mahmood and Aljubury, 2022 [76]	Hybrid Photovoltaic Evaporative Cooling (PV/EC) System Utilizing a Cellulose Pad	Active	600–1200	Water	Up to 20 °C decrease in PV module's temperature Up to 11.2% enchantment in PV module's efficiency
Zizak et al., 2022 [77]	EC Using Cotton-Cellulose Fiber	Passive	0–1000	Water	Up to 20 °C decrease in PV panel's temperature Up to 9.6% increase in PV panel's electrical production
Arkar et al., 2022 [78]	EC Using a Bio-Based Cotton-Cellulose Fiber Later	Passive	–	Water	Up to 25.5 °C reduction in virtual PV cell's temperature
Altegoer et al., 2022 [79]	EC in a Back-Mounted Chimney-like Channel	Active	–	Water	A maximum temperature reduction of 17.5 °C A 8.7% increase in PV module's power output.
Alktranee and Bencs, 2022 [80]	EC System Utilizing Cotton Wicks Immersed into Water	Passive	300–1200	Water	A 22% reduction in temperature of PV panel A 53% improvement in PV panel's efficiency
Alktranee and Bencs, 2023 [81]	EC System Utilizing Cotton Wicks Immersed into Pure Water	Passive	400–1200	Water	A 22.3% reduction in temperature of PV panel A 73% improvement in PV panel's power output PV panel efficiency can be increased by around 97.6%.
Jirapongphatai et al., 2023 [82]	EC Using Hydrophilic Pad Made of Mixture of Polypropylene and Synthetic Fiber	Active	800–1200	Water	Approximately a 46% temperature reduction when the module's temperature is higher than 70 °C, resulting in a 10% increase in electrical production.
Kaneessamkandi et al., 2023 [83]	EC System Utilizing a Heat Pipe Thermosiphon with Acetone	Passive	600–1100	Acetone	Sustain the module temperature in the optimum range.

cooling capacity for the systems [53]. EC systems have been designed for PVs based on evaporation of coolant [57]. The coolant produces lower temperature and higher humidity inside a space during the evaporation. The energy required to evaporate the coolant is provided from the air by cooling it and raising its humidity [53,61]. Systems can be passive or active according to whether they involve any energy input [57]. EC systems are relatively inexpensive, easy to apply and highly efficient techniques, particularly in arid climates [55]. Cuce and Rifat [53] categorized the evaporative systems based on direct contact or indirect contact regarding interaction between the streams, shown in Fig. 6.

The direct EC systems are suitable for hot and arid weather conditions and effectively lower the temperature of the incoming air.

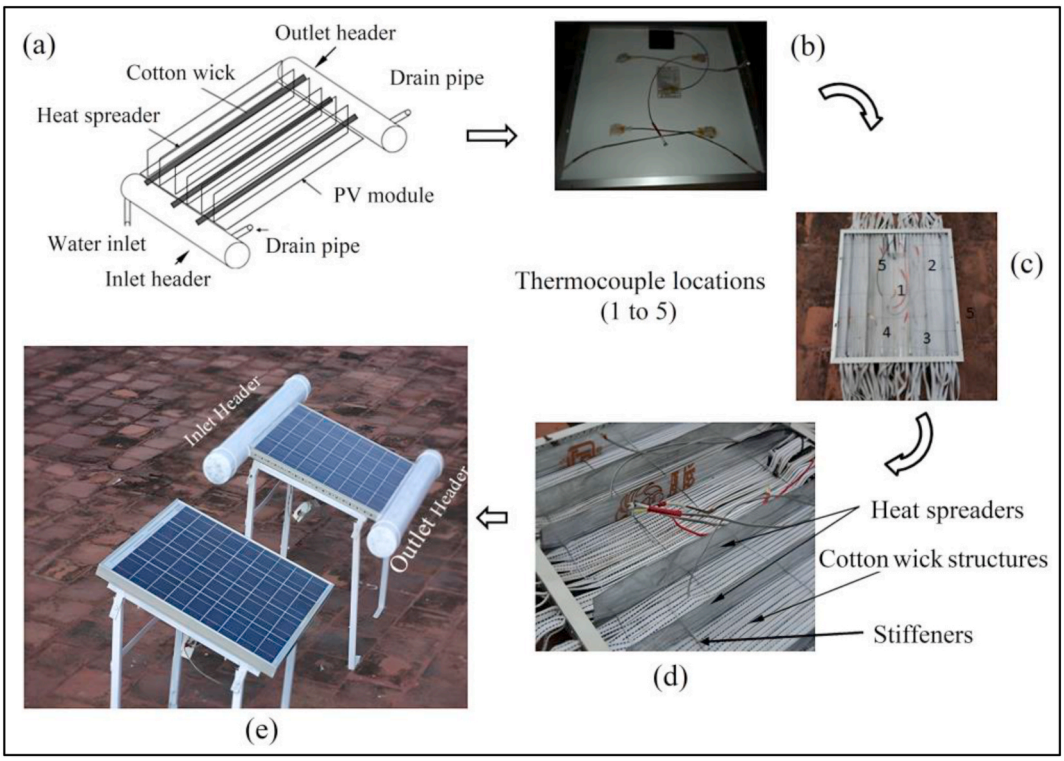


Fig. 7. A visual representation of the experimental setup with fabrication steps. (a) The back surface of the PV module; (b) Thermocouples' location; (c) The spreaders in combination with cotton wick structures; (d) Details of the components; e) The completed experimental setup [63].

However, the cooling performance of an indirect EC system is significantly lower compared to the direct contact system [53].

2. EC systems for PV modules

Experimental, numerical, theoretical studies related to EC systems for PV modules are presented in this review study. It aims to inform researchers about current techniques that have been used for EC systems for PV modules. Thus, it may assist and inspire them to develop new methods and systems, and to save the time for literature study.

2.1. Experimental studies

Experimental studies have been carried out to examine the effectiveness of employing EC techniques on PV modules. The objective of these studies is to enhance the overall performance and efficiency of PV systems by lowering the modules' temperature and diminishing the adverse impacts of temperature evaluation on PV cell performance.

In this section, the experimental studies were titled based on the material and techniques used in their works. The experimental studies mentioned in this review are summarized in Table 1. They are listed at the table based on their publication dates.

2.1.1. EC system utilizing synthetic clay

Alami [57] developed a passive EC system using synthetic clay with different thicknesses to control the temperature increase in PV modules, and to examine the impact of the cooling on the efficiency of the modules. The cooling system basically consist of copper cooling components covered in synthetic clay. A cooling component was positioned to the back of the modules. It provided a thin layer of water to evaporate, thus allowed indirect EC, and decreased the module's temperature. In order to control the amount of evaporation and increase the heat transfer, the pores are made on the clay layer. The thinner clay layer demonstrated the lower enclosure temperature, so the copper sheet was more effectively exposed to natural convection, resulting in greater drop in the module's temperature. It was claimed that during the peak solar period at noon, 40 °C reduction in the module's temperature could be reached. There was an average enhancement of 15% in the output power and an average 16.7% increase in the output voltage during solar noon. However, the study did not provide specific information on the actual variation of panel temperatures over time. The cooling system was indicated to be inexpensive and environmentally friendly. It did not require any energy input, therefore; the electrical energy produced from PV modules was not consumed for cooling the modules. However, while the presented passive EC systems offers promising results in terms of temperature reduction and efficiency improvement, addressing their limitations and drawbacks will be crucial for its successful deployment in real-world solar energy applications. Further research, including field trials and long-term studies, can provide valuable insights into the system's performance, durability, and scalability in different real-world conditions.

2.1.2. EC systems using cotton wicks

Chandrasekar et al. [62] created a passive EC system with cotton wicks structures in combination with water. The cotton wick structures were positioned at the rear side of a PV module in circular ring style and the cotton wicks' end were dipped in water stored in a reservoir. The cotton wicks absorbed and carried water to the module's backside.

There was a 30% decrease in module temperature. A peak module efficiency of 10.4% was achieved by employing the EC system. Furthermore, Chandrasekar et al. [62] used nanofluids (Copper Oxide/water, and Aluminum Oxide/water) in conjunction with the cotton wicks, which were not effective like water in terms of temperature reduction.

Chandrasekar and Senthilkumar [63] developed a passive EC system utilizing aluminum heat spreaders in combination with cotton wicks to control the temperature of PV modules. The spreaders in conjunction with cotton wicks were located at the back surface of a PV module, and a reference PV module without cooling was used for the sake of comparison. The cotton wicks' ends were submerged in water stored in the

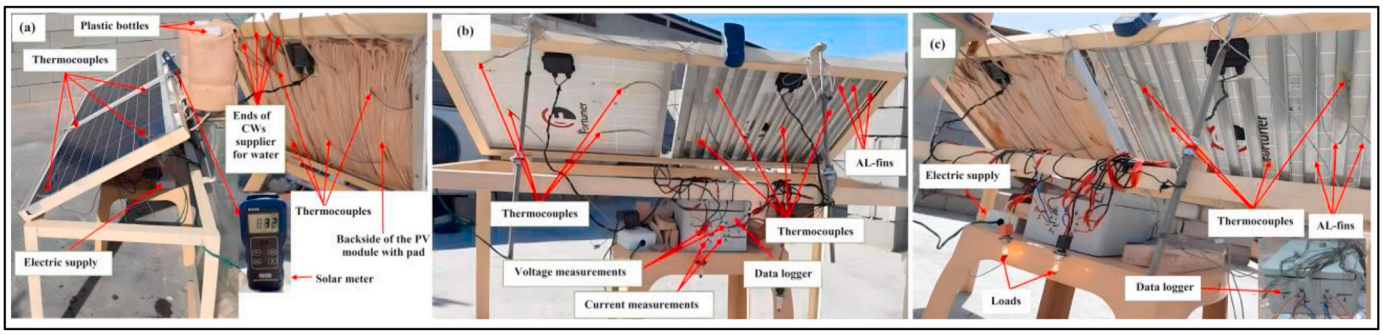


Fig. 8. (a) Experimental setup of the EC; (b) Experimental setup of RAFs cooling; (c) Experimental setup of the evaporative and RAFs cooling together [81].

Table 2
Experimental findings of the evaporative and RAFs cooling.

Cooling technique	Temperature reduction (%)	Improvement in power output (%)	PV panel's efficiency increase
EC with cotton wicks	22.3	73	97.6
RAFs cooling	6.7	13.8	34.2

headers. The aluminum heat spreaders were designed and fabricated in the study. Thermocouples were positioned at the rear side of the PV modules at five different locations to measure the modules' temperature. Also, to ensure effective contact between the cotton wicks and the spreaders, as well as prevent the cotton wicks from sagging, aluminum spikes were utilized as stiffeners. A visual representation of the experimental setup with fabrication steps was displayed in Fig. 7.

It was concluded that because of the cooling effect delivered by the evaporation cooling and the fin effect of the heat spreaders, there was a 12% reduction in PV module temperature, leading to 14% increase in electrical power output of the module.

Alktranee and Bencs [80] studied an EC technique using cotton wicks immersed into water that covered the back side of the PV panel. The cotton wicks' ends were submerged in plastic water bottles to absorb and move water to the panel's backside. The cotton wicks transferred the water fast and evenly distributed it over the backside of the panel because they have a great absorbent capacity. Also, they did not have a reaction with the rear layer of the PV panel after being submerging in the water [80,81]. Alktranee and Bencs [80] indicated that there was a 22% reduction in the panel temperature, creating a 53% improvement in panel's efficiency. Furthermore, Alktranee and Bencs [81] repeated this EC technique with pure water rather than regular water. The experimental findings stated that there was a similar temperature reduction of 22.3%, resulting a 73% improvement in the power output.

It was noted that the cotton wicks were easy to obtain, do not demand maintenance. Also, they indicated that even though using pure water led to additional expenses in the system, it was essential because salty water decreases the absorption capacity of the cotton wicks due to the adherence of salt [81].

In addition, Alktranee and Bencs [81] compared the EC system utilizing cotton wicks to a PV panel equipped and thus cooled by rectangular aluminum fins (RAFs), and a reference PV panel without cooling. Fig. 8(a), (b), and (c), show a panel with the EC, a module with RAFs, and panels with both the EC, and RAFs.

The results from the EC and RAFs cooling are shown in Table 2 below.

Based on the experimental results, it can be said that the EC had a better potential than RAFs cooling.

Additionally, Chandan et al. [74] experimentally compared the wick cooling to active air-cooling with low and high flow rates, and active

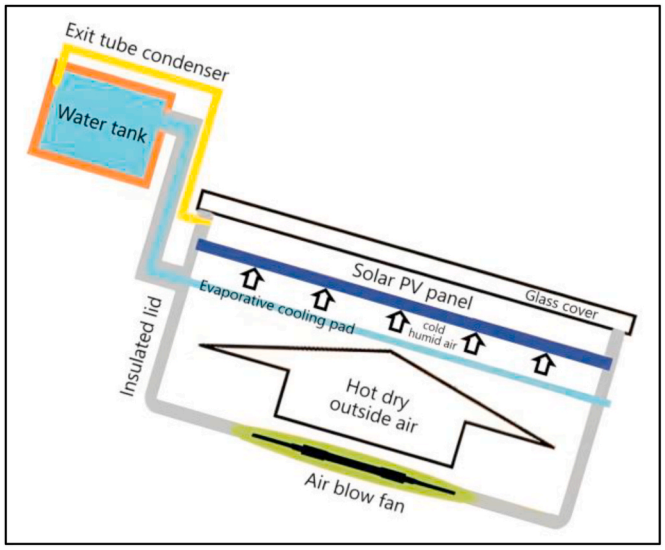


Fig. 9. Illustration of the EC system [56].

water-cooling in outdoor and indoor conditions. Although the active cooling techniques were more effective than the wick cooling in terms of temperature reduction, the wick cooling was a viable option considering the complexities and expenses associated with the active cooling systems.

The studies by Chandrasekar et al. [62], Chandrasekar and Senthilkumar [63], and Alktranee and Bencs [81] collectively explore the effectiveness of passive EC systems, specifically employing cotton wicks, in enhancing the performance of PV modules. These cooling approaches leverage the natural evaporation of water to regulate module temperatures and subsequently improve electrical power output.

In conclusion, these studies collectively highlight the potential of passive evaporative cooling systems, particularly those employing cotton wicks, as efficient and cost-effective methods for enhancing the performance of PV modules. The consistent theme across these investigations underscores the importance of careful design and material considerations in optimizing the effectiveness of passive cooling technologies for sustainable solar PV applications. While the studies highlight the benefits of passive evaporative cooling with cotton wicks for PV modules, certain considerations should be kept in mind. These include the potential environmental dependence, questions about long-term durability, scalability challenges, water usage concerns, effectiveness in various climates, material compatibility, energy input for manufacturing, and the need for continuous assessment against these technologies.

2.1.3. EC systems using co-current configurations

In their study, Suresh et al. [56] improved a cost-effective

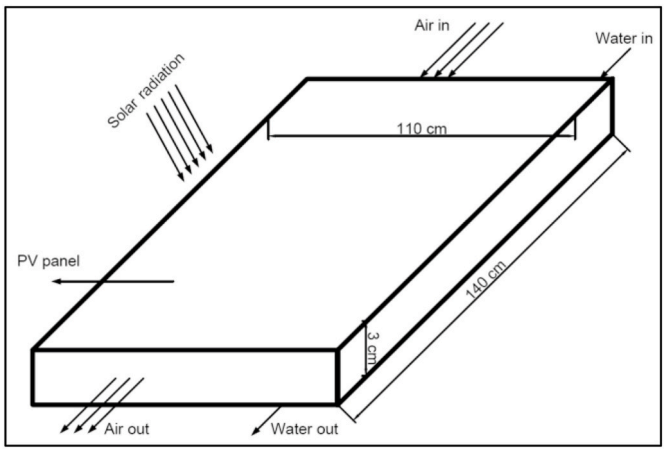


Fig. 10. Demonstration of PV panel with EC operating with co-current configuration [69,84].

evaporative system based on employing capillary action to a wetting pad with water, then blowing air into the pad. This action provided a cooling effect, resulting in cooled air. The cooled air extracted heat from the PV panel' surface by forced convection. Fig. 9 displays the EC device consisting of a sealed chamber equipped with a cooling pad, a water tank, an air blow fan, and a top glass cover to enhance humidity levels.

The study reported the impact of temperature on various parameters such as short circuit current, open circuit voltage, fill factor, and efficiency. It was agreed that even though temperature increase in the PV panel improved the current of the panel, it decreased the voltage of the panel, causing a decrease of the fill factor, thus the efficiency of the panel. Although it was stated that a 10% efficiency improvement was achieved with the cost-efficient evaporative system, the specific details of the experimentation, including data on temperature variation over time, radiation levels, ambient temperature, and module specifications, were not provided.

Haidar et al. [69] conducted an experimental study based on creating co-current configuration to investigate the impact of EC on the performance of PV panels. Co-current flow configuration was created by a fan blowing ambient air inside the duct alongside the water flow. The water flowed over a piece of cloth located on the lower side of the duct which was thermally insulated. The cloth provided a uniform water distribution. Water was sent to the cloth from a water tank through a flow meter by a pump. When air flow inside the duct encountered the wetted cloth, water evaporation began, consequently both water and air temperature reduced, and absorbs the unwanted heat from the rear surface of the panel. Thus, it decreased the panel temperature, leading to higher electricity generation. A reference PV panel without cooling was used for comparison purposes. Thermocouples were located the rear surface of the reference and cooled panels to measure the temperature differences. Fig. 10 displays a schematic diagram for the PV module with the EC.

There was a temperature reduction of over 10 °C, resulting in a 5% enhancement in power output of the PV panel. Furthermore, Haidar et al. [69] claimed that the EC operating with co-current configuration was effective, required a minimal amount of water to provide uniform cooling.

Both studies demonstrate the potential of evaporative cooling in enhancing PV panel performance. They contribute to the understanding of EC systems for PV panel cooling, with Suresh et al. [56] focusing on a cost-effective approach and Haidar et al. [69] emphasizing efficiency through a co-current flow configuration.

2.1.4. PV evaporative chimney

A solar chimney delivers a passive air passage that depends on thermal buoyancy as the primary force for airflow. Due to solar energy, temperature of the air inside the chimney increases, causing density drop in the air. The decrease in the air density leads to the upward movement of the air inside the solar chimney, resulting in its expulsion from the chimney's top. Solar chimneys have various applications such as, building ventilation, drying processes, and electricity generation systems [61].

Lucas et al. [61] proposed an alternative environmentally friendly solar cooling design called Photovoltaic Evaporative Chimney for PVs. The thermal and electrical performance of a PV module cooled by the evaporative chimney was experimentally assessed for varying levels of incident solar radiation and various ambient conditions such as ambient temperature, and relative humidity.

The proposed design has two objectives: firstly, to cool the PV panel, and secondly, to dissipate heat generated by a refrigeration cycle. In order to achieve these goals, the solar chimney was separated into two primary components. The first part is called the EC zone; having a series of nozzles which release water in a parallel direction to the downward airflow. In this zone, there was a process of heat and mass transfer taking place between water and air. When the water went down, a fraction of it evaporated, leading to the cooling of the remaining water. The EC zone operated as a compact cooling tower. The second section called convective zone, is basically considered a solar chimney where the air stream passed through the rear and cooled the PV panel. It resulted in enhanced performance. PV Evaporative Chimney method and air

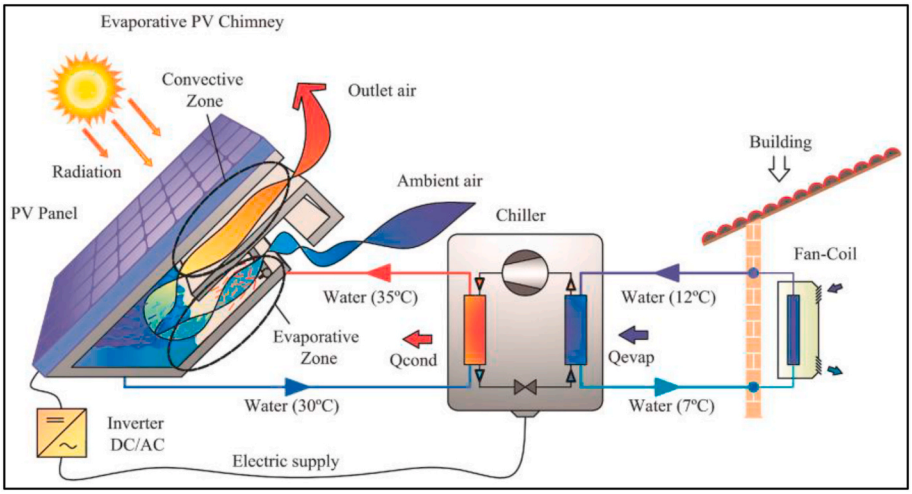


Fig. 11. PV Evaporative Solar Chimney and Air conditioning system [61].

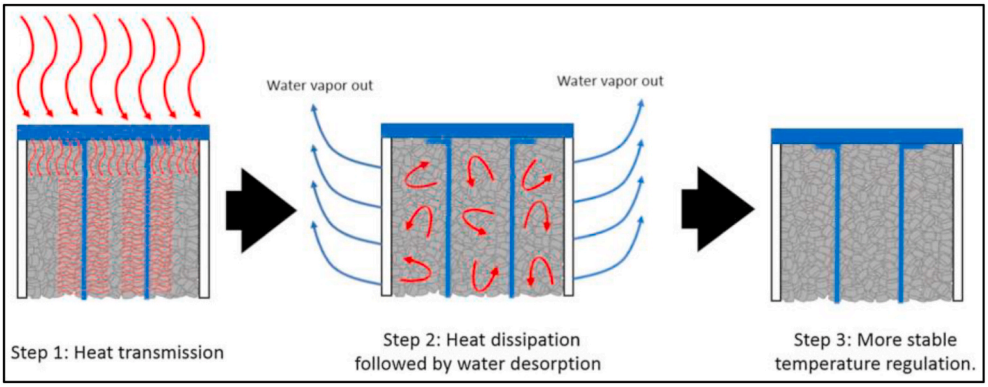


Fig. 12. The working principle of the proposed system [66].

conditioning system is shown in Fig. 11.

It was concluded that there was a temperature decrease of up to 8 °C and electrical efficiency improvement ranging from 4.9% to 7.9%.

Additionally, Lucas et al. [64] developed a modified version of the evaporative chimney design presented in Lucas et al. [61]. A sliding water system was included in the previous design, which was the most significant modification. In order to slide the water over the chimney panel, a new hydraulic design was made and constructed. Following the water sliding process, the water was gathered from the lower section of the PV panel trough a drainage channel and subsequently returned to the water storage tank.

The modified evaporative chimney system provided an average temperature reduction of 15 °C and a 10% enhancement in electrical efficiency. It demonstrated better results compared to the first design. Furthermore, Ruiz et al. [68] utilized PV panels cooled by PV Evaporative Chimney to power a water-cooled chiller used in the building sector. It was proved that a PV solar-driven cooling system can be improved with the PV Evaporative Chimney, consequently making the system more economically feasible and energy efficient.

The studies by Lucas et al. [61,64] and Ruiz et al. [68] introduce innovative applications of the photovoltaic evaporative chimney, showcasing its versatility in enhancing both the thermal and electrical performance of PV modules. These studies collectively highlight the photovoltaic evaporative chimney as a promising and adaptable technology for improving the performance of PV modules, emphasizing its role in both thermal cooling and electrical efficiency enhancement.

2.1.5. EC system using cloth and rubber pipes

Haidar et al. [55] introduced an EC technique using cloth and rubber pipe attached directly to the back of the cooled PV panel that were left open and exposed to the surrounding air. There was not a fan utilized in the system. The water was fed to the rear of the PV panel from a tank via gravity and wet the cloth using rubber pipes. A valve was located between the tank and pipes to manually control the water flow. The works made sure that the cloth was tight and contacted the panel back surface to avoid any gab formation between the PV module and cloth. In this system, the cloth is in direct contact with the back surface of the panel, when it becomes wet and creates a difference between the water vapor concentration at the water-air interface (known as the saturation layer) and the concentration of water vapor in the surrounding dry air, it absorbs heat from the module, causing a decrease in PV panels' temperature. Haidar et al. [55] revealed that there was a remarkable reduction of over 20 °C in the temperature of the PV panel, as well as an approximate 14% increase in electrical power production in comparison to the reference panel. This low-tech approach demonstrates the potential for efficient PV panel cooling without the need for additional fans or complex systems.

2.1.6. EC system with wood wool and heat sink

Hasan et al. [65] designed a passive cooling system comprising of a wood wool material combined with heat sink structures made of aluminum alloy for PVs. The system was designed according to the EC concept and enlarged surface area of hear transfer. Pin fins were modeled and utilized as heat sink to extend the surface area of heat transfer. It was ensured that the moist wool wood pad completely touched the surface area of the heat sink.

A continuous water supply was provided to maintain the desired moisture level of the wood wool pad. The mass flow rate of water was controlled using a fixed regulated valve positioned inside the supply hose. To absorb the unwanted heat on the panel, the water was steadily dropped onto the pad through a perforated pipe.

The works reported improvements in the PV panel's performance, including a 32.7% increase in the panel's output power and a 31.5% improvement in module efficiency compared to the reference panel without an EC system with a heat sink. The cooling system decreased the panel temperature around 27%.

2.1.7. EC system using zeolite

Zeolite, also called activated alumina, is a promising water absorbent material. It stores water in the pores of its inner particles and prevents it from leaking or evaporating without significant heating effects. It has an ability to adsorb water up to 15% of its weight at normal atmospheric temperature and allow the water to evaporate when the temperature increases. Also, it is cheap to be manufactured and can be found in nature [66].

Abdallah et al. [66] declared a novel evaporative system utilizing the porosity and water adsorption characteristic of zeolite to effectively lower the temperature of PV panels. When a certain amount of water-saturated zeolite located at rear surface of a PV panel receives a sufficient amount of heat, the zeolite releases water in the form water vapor, thus removes heat from the panel. The proposed cooling process utilizing zeolite is shown in Fig. 12.

Instead of using actual PV panels, in studies blue coated copper plates were used having similar features of a PV panel. The experiments were conducted indoors, and solar lights was simulated by halogen lamps. Varying solar radiations of 600, 800, and 1000 Watts per square meter (W/m^2), and different system configurations (including fins, and using metal mesh) were employed during the experiments. An uncooled plate was used for comparison purposes. In conclusion, depending on radiation intensity, the temperature of solar system with the EC could be maintained at a range of 9 °C–14.9 °C. As a result of temperature reduction, an average improvement ranged from 6.5% to 10% could be received. The utilization of zeolite in this cooling system presents a cost-effective and nature-friendly solution for enhancing the performance of PV panels, showcasing its potential for practical applications in the field of solar energy. The study underscores the importance of exploring alternative cooling methods to optimize the efficiency of solar energy

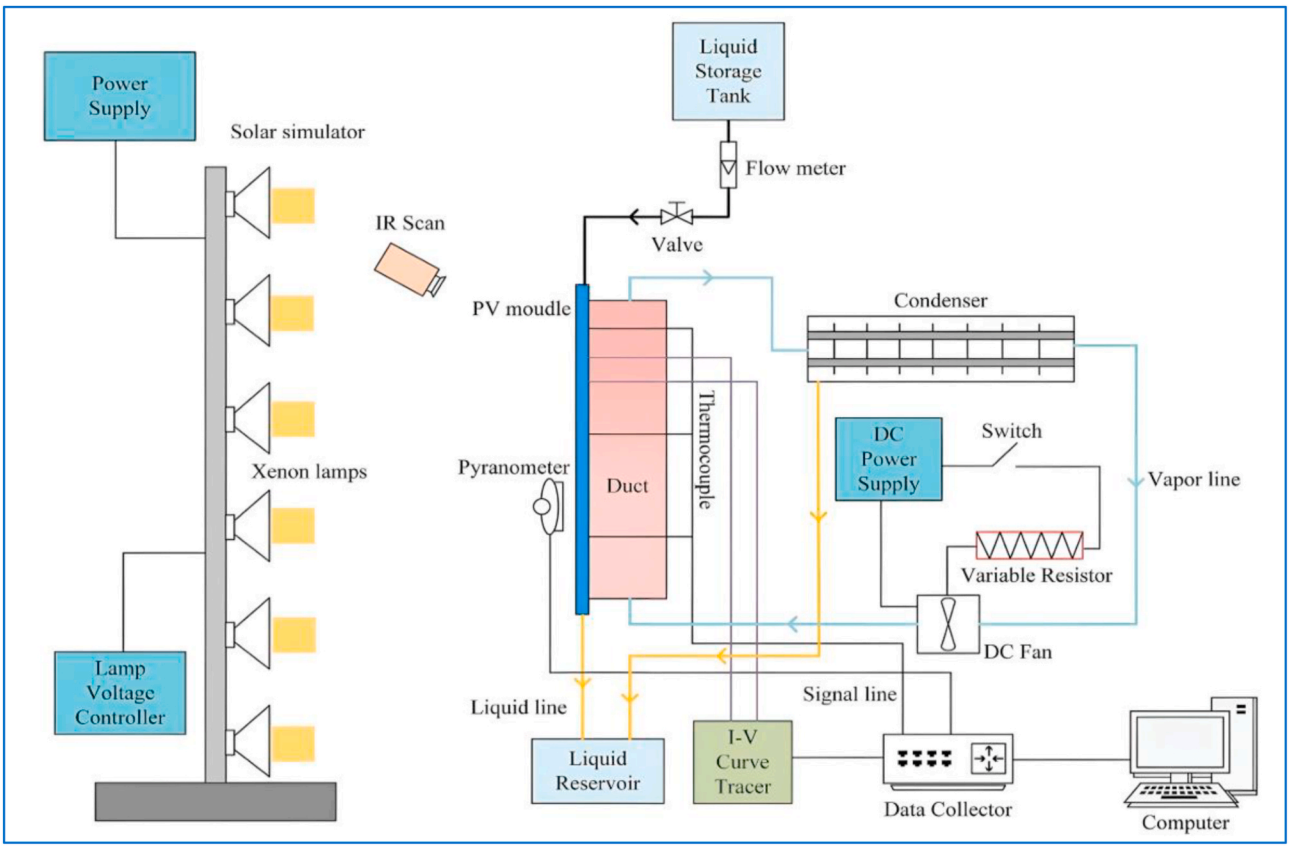


Fig. 13. Schematic illustration of the experimental setup [67].

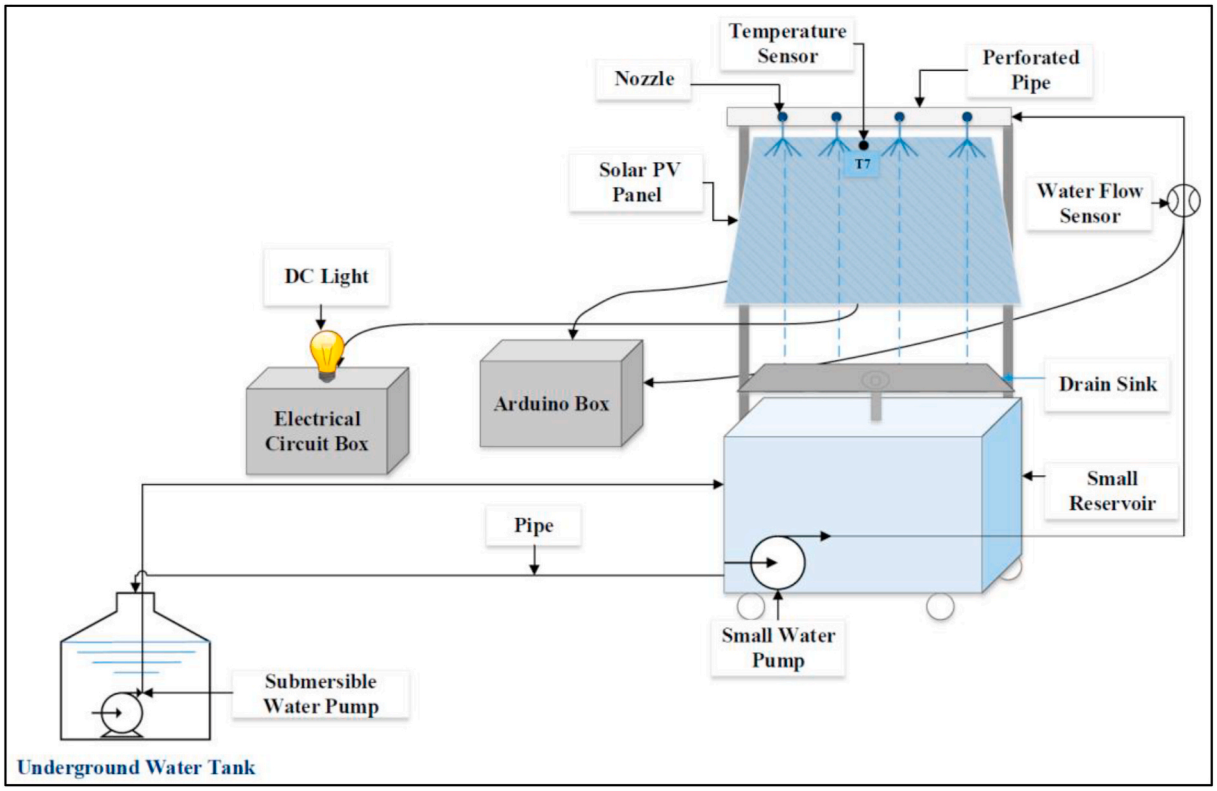


Fig. 14. Front sketch of the cooling system [70].

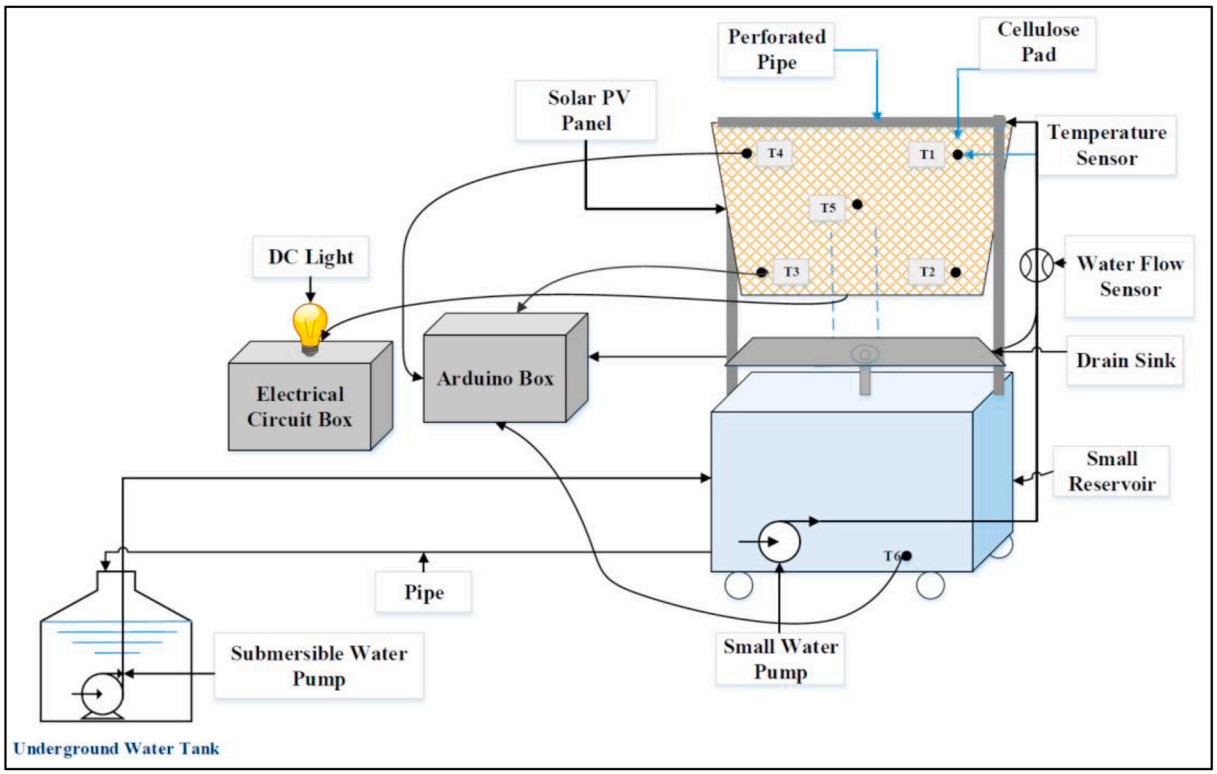


Fig. 15. Back sketch of the cooling system [70].

systems. However, while the zeolite-based evaporative cooling system presents a promising solution for improving PV panel efficiency, addressing its drawbacks such as scaling for larger systems and dependency on ambient conditions are essential for its practical implementation and widespread adoption in real-world PV applications.

2.1.8. EC cooling system with porous layer

Wang et al. [67] designed and fabricated an active phase change cooling system based on porous media applied in PV modules. The method aimed to manage thermal and efficiency of the modules. Ethanol was utilized as coolant. A liquid distributor was designed and used to ensure a uniform distribution of coolant in the porous layer positioned at the back side of a PV module. The liquid coolant was supplied by gravity to the distributor from a storage tank located above the module, maintaining a constant inlet temperature. Even distribution of ethanol within the porous layer played a key role in regulating the temperature of the module. When the temperature of the module increased, the coolant absorbed the excess heat by evaporating. The remain unevaporated ethanol was then flowed into the liquid reservoir. The ethanol gas was propelled toward a condenser by a fan. The condensed ethanol directed back into the reservoir, so it was recovered and resent to the cooling system. Needed solar irradiation was provided by an artificial solar simulator. A schematic illustration of the experimental setup is presented in Fig. 13.

In conclusion, it was stated that there was a significant improvement in the electrical power output and efficiency of the PV module by implementing the active phase change cooling system. Under an irradiation of 900 W/m^2, the electrical power output and efficiency improvements were 19.32% and 18.40%, respectively. The highest specific power enhancement was 21.37 W/m^2. This presented cooling mechanism effectively manages the thermal conditions, preventing overheating and improving overall module efficiency. This study underscores the efficacy of the active phase change cooling system in optimizing PV module performance, offering a promising solution for advancing PV energy technology.

2.1.9. EC system utilizing water from an underground tank

Kadhim and Aljubury [70] utilized water from a buried underground tank as coolant for PVs. The buried underground tank maintained the cooling water temperature at around 18.5 °C, which was not higher than the daily ambient temperature. This underground water was then stored in a small reservoir positioned under the PV panel. The cellulose pad was located on the rear surface of the panel for EC and nozzles were placed on the front side for spray cooling, and the panel's surface cleaning. The pad was saturated with water from the reservoir. This water was delivered to the pad through a perforated pipe located at the top of the panel's back surface by a small water pump. The sketch of the front and back views of the cooling system are displayed in Figs. 14 and 15, respectively.

The studies conducted experiments for different cases. The first case considered only backside EC which demonstrated a 1.74% improvement in electrical efficiency and an average 4 °C drop in the temperature of the panel. In the second scenario, a combination of backside EC and frontside spray cooling were employed, and the panels' temperature reduced twice more. A 1.06% higher electrical efficiency was received compared to the first case. Also, different water flow rates were used to receive the best results.

The utilization of underground water resources aligns with sustainable practices, and the experimental results provide a foundation for further exploration and optimization of such integrated cooling strategies in the field of PV technologies.

2.1.10. EC system with an atmospheric water harvester

Li et al. [71] designed an EC system for commercial PV modules based on a sorption-based atmospheric water harvester to cool the modules and thus increase the modules' power output. They designed a hydrogel-based atmospheric water harvester connected directly to the rear side of a commercial PV module. The harvester comprised of a layer (PAM-CNT-$CaCl_2$) made of cross-linked polyacrylamide (PAM) implanted with carbon nanotube (CNT) and calcium chloride ($CaCl_2$) as water vapor sorbent. It gathered and retained a significant amount of

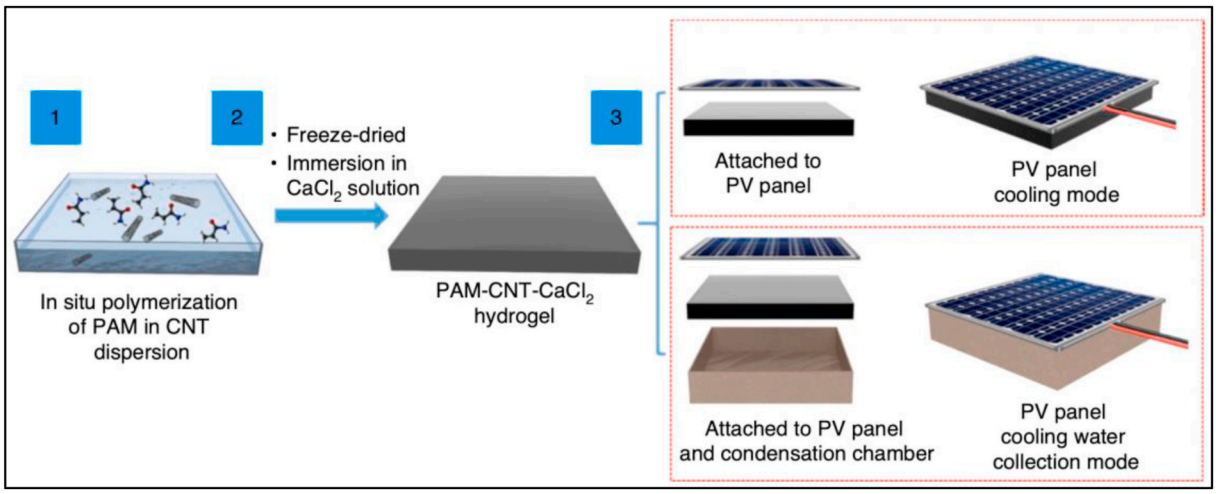


Fig. 16. The illustration of the process of producing the hydrogel and the operational modes: PV panel cooling mode, and cooling water collection mode [71].

atmospheric water in the evening and night, and then released it during the day as water vapor by utilizing the heat produced by a PV module. Consequently, in in laboratory conditions, the cooling system decreased the module temperature by at least 10 °C when exposed to 1000 W/m^2 solar irradiation. Also, during the outdoor tests carried out in winter and summer, the cooling system displayed a significant improvement of 13–19% in electricity production for the panel. The process of manufacturing the hydrogel and the operational modes is illustrated in Fig. 16.

Li et al. [71] claimed that this cooling system can be employed various type of PV systems, ranging from a single individual household PV panel to industrial PV farms.

The integration of a sorption-based atmospheric water harvester with PV technology represents a good approach to address both cooling and water resource challenges, presenting a potential breakthrough in sustainable and efficient energy production. This approach not only contributes to advancing the field of solar energy but also underscores the interdisciplinary potential of combining technologies for enhanced environmental and energy solutions.

2.1.11. EC systems using fabric and burlap cloth

Dida et al. [72] developed a passive EC system using the burlap cloth's capillary action to reduce the excessive heat produced by a PV module, aiming to increase its overall efficiency. The cloth was directly connected to the back of the module. The top part of the cloth was submerged into the water supply basin filled by a 50-L (L) water tank located above the module. Hence, the cloth transferred the water to the rear side of the module by the capillary action and with assist of gravity force. The water not being absorbed by the cloth was recovered from a small hole in the bottom the module with help of an elastic hose and a small water tank. A reference module was used for comparison purposes. The results of the experiment demonstrated that the module temperature was lowered by 20 °C (equivalent to 26% reduction), resulting in a significant boost of 14.75% in the electrical efficiency. The results further stated that the cooling system provided a homogeneous temperature distribution over the module during the entire day. Moreover, Dida et al. [72] indicated that the cooling system is environmentally friendly, inexpensive, noiseless, and does not require large amount of coolant.

Almuwailhi and Zeitoun [73] installed an insulated channel on the back of a polycrystalline PV panel to investigate the effect of natural convection, forced convection, evaporative natural convection, and evaporative forced convection on energy production and efficiency of the panel. A wetted fabric was employed as the evaporation material to facilitate evaporative natural convection cooling along the cooling

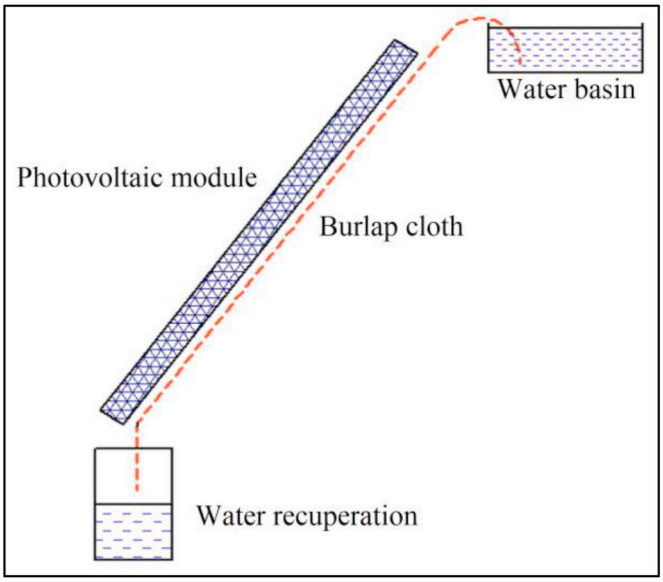


Fig. 17. The flow mechanism assisted by gravity [75].

channel, while evaporative forced convection cooling was achieved by fans blowing air over the wetted lower surface of the channel. Natural convection cooling was more effective with an increase width of the cooling channel. Natural convection increased the panel's efficiency and energy production by 1.7%, and 1.2%, respectively, with using an air gab of 120 mm. Enhancements in forced convection cooling were observed with an increase in air velocity. In forced convection cooling, utilizing air at a speed of 3 m/s improved the efficiency and daily energy production of the panel by 4%, and 4.4%, respectively. Relative to natural convection cooling, evaporative natural cooling was more effective. There was an increase of 3.6% in energy output and an improvement of 2.7% in efficiency of the panel for the case of using evaporative natural cooling. In comparison with forced convection cooling, evaporative forced cooling improved the cooling and boosted the efficiency by 3.8%, and daily energy production by 3.8% at a speed of 2 m/s.

Malvika et al. [75] also used the gravity-assisted EC system using burlap fabric layer attached to the rear side of a PV module. Single- and double-layer arrangements of burlap fabric was utilized. Also, in another case called anti-gravity mode, the bottom of the burlap fiber was immersed into a water tank located below the module; therefore, water

Table 3
Experimental results of hybrid PV/EC system.

Technique	Increase in electrical efficiency (%)	Temperature drop (°C)
Gravity assisted flow with single layer	3–18	15.6
Gravity assisted flow with double layer	18–21	21.1
Burlap layer in anti-gravity mode	13	8.7
Phase change material	2	8.1

had to move against gravity to the rear side of the module owing to the capillary action. Fig. 17 displays a visual presentation of the flow mechanism used in the gravity-assisted EC.

The experimental results from the EC systems were compared to the phase change material cooling conducted in the same environment. It was discovered that both EC configurations outperformed the phase change material cooling. Table 3 presents the experimental results for all cases used in the study. A reference module without cooling was used for the purpose of comparison.

2.1.12. EC systems using cellulose pads/fibers

Mahmood and Aljubury [76] invented a hybrid photovoltaic evaporative cooling (PV/EC) method utilizing a cellulose pad and a water distribution pipe attached to the back of the cooled PV module. A cross-flow centrifugal fan, and a small direct current submersible water pump were operated for air flow and water circulation, respectively. The water pipe was used to provide a uniform water distribution on the pad, shown in Fig. 18(a). The fan was located at the top end of the duct channel to effectively draw out air from the interring channel through the cellulose pad, and then expel it outside the duct, demonstrated in Fig. 18(b). In this study, the experiments were conducted for the cellulose pad having a varying thickness (50, 100, and 150 mm) with different water flow rates (1,2, and 3 L per minute). Air velocity was kept between 2 and 3 m/s. Fig. 18(c) displays schematic diagram of the experimental setup.

Experimental results of this study are presented in Table 4. Increase in the pad thickness improved the panel's electrical performance and reduced its temperature further.

Zizak et al. [77] presented a passive EC method using a porous

Table 4
Experimental results of hybrid PV/EC system.

Thickness of the cellulose pad (mm)	Increase in the PV panel efficiency (%)	Reduction in the temperature of the PV panel (°C)
50	7.4	12
100	10.5	20
150	11.2	20

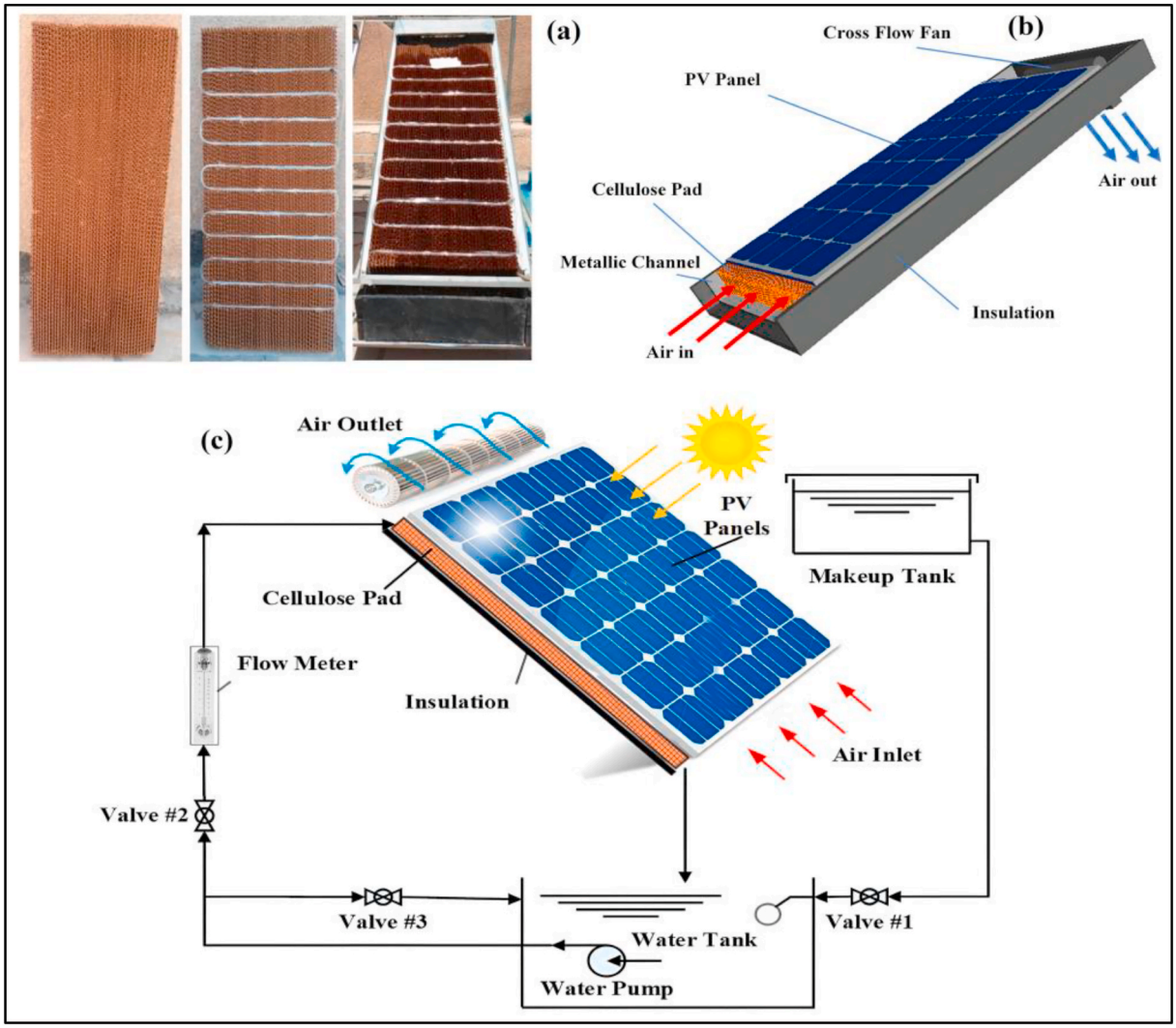


Fig. 18. (a) Setup of the cellulose pad and water distribution pipe; (b) PV/EC system schematic layout; (c) Illustration of experimental setup [76].

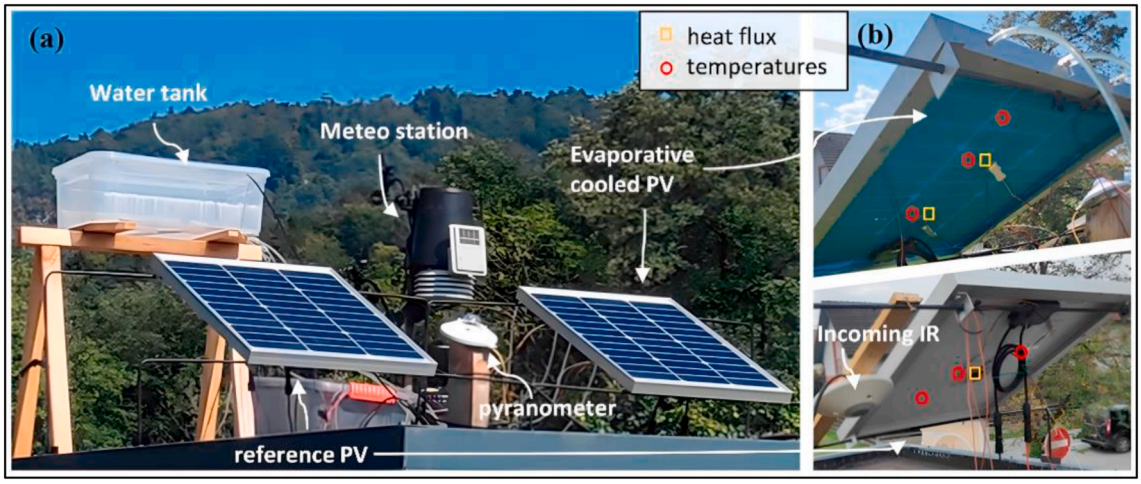


Fig. 19. (a) Demonstration of the experimental setup on the rooftop; (b) Back view of the cooled PV panel (on the top) and reference panel (on the bottom) [77].

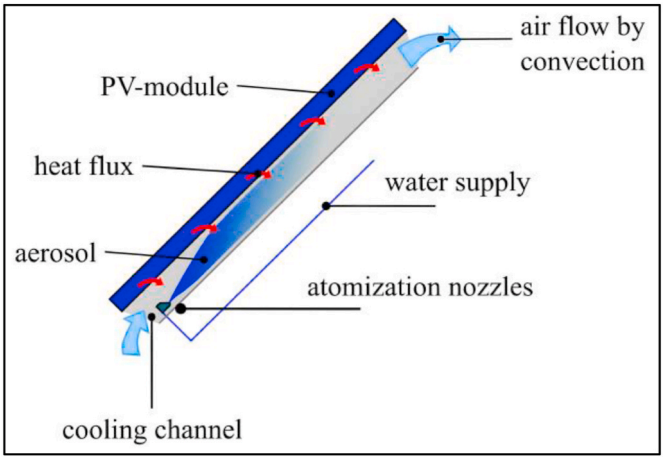


Fig. 20. The schematic presentation of the cooling system [79].

cotton-cellulose fiber evaporative layer as a cooling system fixed to the back surface of the PV panel. Water as a coolant was sent from a tank positioned above the panel to the evaporative layer through four tubes uniformly distributed across the width of the panel by gravity. This method did not require a pumping system. Experimental setup of the EC system with a back view of the panels is shown in Fig. 19(a), and (b).

In order to maintain a constant saturated evaporation layer, the water outflow was adjusted by observation. The amount of droplets discharged at the bottom of the panel was not more than five per minute. As a result, compared to the panel without the EC, a maximum 20.1 °C of temperature decrease was received, leading to a 9.8% power increase in electrical generation of the PV panel. Moreover, to reveal whether this EC system was sustainable or not for an area, studies determined the needed rainwater volume and roof size.

Arkar et al. [78] attached a bio-based cellulose fiber layer at the rear side of a virtual PV cell to deliver an EC. The fiber is sustainable and has a remarkable water absorption and distribution properties. It could absorb and uniformly distributed 54 mL of water over the surface of the cell in 11 min. Consequently, the EC system decreased the cell temperature up to 25.5 °C. The specific details of the experimentation were not presented in the paper.

2.1.13. Water droplet evaporation in a chimney-like channel

Altegoer et al. [79] improved a technique based on droplet evaporation within a cooling channel resembling a chimney. The channel was positioned at the back surface of the PV panel. The sketch of the cooling

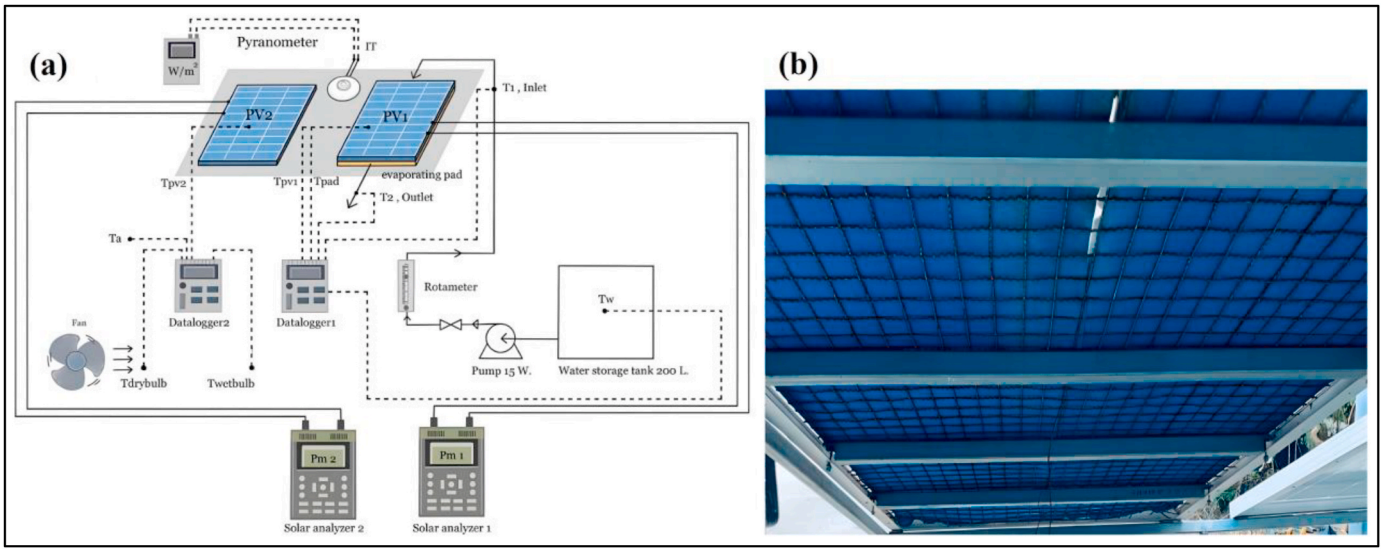


Fig. 21. (a) Experimental setup; (b) Rear surface of the panel with evaporative pad [82].

system is shown in Fig. 20.

Water drops were generated from an uninterrupted stream of water by the optimal atomizer nozzles that were selected with shadowgraph imaging of different atomizer nozzles. The outlet of the nozzle was aimed towards the cooling channel to allow the airborne droplets to evaporate and, consequently extract the heat from the surrounding air. The nozzles were installed in different positions, and different operating parameters were used to reveal the best temperature reduction, power output efficiency, and efficiency increase per water usage.

The best results in water usage per efficiency improvement were received from two pin-jet nozzles located at the inlet and outlet of the cooling channel. Furthermore, the best temperature reduction (17.5 °C) and power output efficiency (8.7%) were attained with the utilization of three hollow cone nozzles attached to the channel inlet.

Moreover, the surface area of airborne water droplets was enhanced by nozzle atomization. When the droplets were smaller in size, they evaporated at a faster rate, extracting more energy from the surrounding air. Conversely, larger droplets withdrew the cooling channel without complete evaporation. This incomplete evaporation resulted in more wasted water without any additional cooling benefits. As a consequence, it was favorable to minimize the size of droplets through atomization to guarantee complete evaporation and less water usage in the cooling channel. Less water waste makes the technique more sustainable and profitable.

2.1.14. EC system using hydrophilic pad

Jirapongphatai et al. [82] introduced a water EC system utilizing a hydrophilic pad made of a mixture of polypropylene and synthetic fiber. The pad was attached at the back surface of a PV module, and the performance of the module was experimentally examined. The coolant was sent to the pad at different flow rates and inlet temperatures from a water storage tank by a small pump. There was a reference module without cooling for comparison purposes. Schematic presentation of experimental setup, and rear surface of the PV panel covered with evaporation pad are illustrated in Fig. 21(a)–and b), respectively.

In conclusion, in the paper, it was expressed that expressed that an approximately 46% temperature reduction could be reached when the temperature of the panel is higher than 70 °C. This temperature drop resulted in a 10% growth in the total electrical production. Additionally, a greater electrical power was received from a higher flow rate and lower inlet water temperature. Compared to the reference module, there was a 6.5%–11.1% growth in net electrical power when the inlet coolant temperature ranged from 20 °C to 30 °C.

2.1.15. ECs utilizing a heat pipe thermosiphon

Kaneesamkandi et al. [83] designed a passive EC system utilizing a heat pipe thermosiphon, closed-loop evaporative system, for PV modules. The system aims to sustain a suitable operating temperature range in a PV module. Acetone was employed as a phase change liquid because it has a favorable boiling point temperature of 56 °C. The heat pipe thermosiphon was attached at the rear side of the module. An additional PV module without cooling system was used for comparison purposes. When the module temperature increased over the phase change temperature, acetone underwent evaporation, causing vapor to ascend along the tube toward the condenser region. The condenser region facilitated the condensation of vapor on its walls by releasing heat to the surrounding. Subsequently, the condensed acetone descended back to the evaporative zone, allowing the cycle to repeat, thus ensuring the desired temperature of the module was sustained.

2.2. Numerical and theoretical studies

Limited numerical and theoretical studies have been conducted to investigate the effectiveness of EC techniques applied to PV modules. The studies have developed mathematical, thermal, multi-parametric, and simulation models to numerically examine the thermal behavior of evaporatively cooled PV modules and reveal the possible benefits of EC systems in improving the overall performance of PV modules. The studies also aimed to confirm the experimental scenarios and reveal the parameters affecting the EC systems for PV modules.

Lucas et al. [61] presented the starting process of modeling the electrical efficiency of PV systems by employing the traditional linear expression, and modeling of the EC process by incorporating efficiency and the number of transfer units (NTU) technique. The correlation between the temperature of the PV module and the electrical efficiency obtained from the experimental data, presuming the traditional linear expression for PV electrical efficiency. The effects of the independent variable such as the difference between ambient and intermediate air temperature, difference between the air velocity and wind speed, on the efficiency of both PV model with the EC cooling and the one without the cooling was exhibited by utilizing efficiency data from the experiments and the NTU method. It was claimed that further analysis was left for the future work.

Suresh et al. [85] carried out numerical and experimental investigation to reveal the impact of EC cooling employing an evaporative cooling pad attached at the back side of solar PV module. In this study, a two-dimensional steady-state model was developed for the cross-flow interaction between water and air to reveal the temperature, and absolute humidity profile of air through the cooling pad for different inlet conditions of air and water. The model involved a set of ordinary differential equations solved by the Runge-Kutta method with a changing time step. A computer code created in MATLAB was used to acquire the results. As a result, in every case, humidity experienced an increase while air temperature declined across the cooling pad due to latent heat addition. Thus, the study validated the latent heat effects and humidification process.

Gleebpratum and Salakij [86] introduced thermal models to forecast the temperature on the rear surface of PV panels with a Salo fabric EC and PV panels without cooling. The models were developed according to the energy equation in the context of heat conduction. To validate the thermal models, experiments were carried out on a PV model constructed from aluminum plate equipped with a silicone heater to simulate a real PV panel. Also, a cooling device was attached one of the plates to simulate the cooling effect. The thermal models nearly anticipated the rear side temperature of the PV model when compared to the experimental data.

Abed et al. [87] developed a numerical simulation model to examine the improvement in performance of a PV water cooling module. The module used a passive cooling method utilizing a wetted cotton porous wick fixed to the back surface of the PV panel. A panel without cooling was used as a reference panel. A COMSOL software Ver.5.3. was used to model and simulate equations of the system based on time-dependent energy balance. The data for input parameters such as solar radiation, ambient temperature, wind speed, water inlet temperature, were obtained from a previous experimental study. In conclusion, there was an approximately 9.57% reduction in the module's back side temperature, resulting about 10.59% increase in the electrical power of the panel because of the EC cooling. Moreover, the model proved that the temperature of the PV panel increased when solar radiation intensified.

Poddar et al. [30] used mathematical relations and Ansys software to determine surface temperature and the performance evaluation of a PV module system. Evaporative cooling and thermoelectric modules were employed as the adject to the PV system to improve the power output and overall efficiency. The study also revealed the effects of process parameters such as temperature coefficient, irradiance, concentration ratio, and wind speed, on the performance of the PV system. The cooling impact provided by EC module lowered the temperature of the PV system combined with thermoelectric module, significantly increasing its efficiency compared to both the PV module and PV- thermoelectric module hybrid system. A rise in solar concentration ratio elevated the overall temperature of all hybrid module systems and could diminish the power output from the systems. However, the temperature increase

Table 5
Numerical and theoretical studies on EC of PV modules.

Study	System description	Model	Method	Coolant	Findings
Lucas et al. [61]	PV Evaporative Chimney	Numerical	Traditional linear expression, and NTU	Water	Provided the initial steps of modeling the PV electrical efficiency. Presented the effects of independent variable on the efficiency.
Suresh et al. [85]	EC System with Cooling Pad and Air Blow	Numerical	The Runge-Kutta method, and MATLAB	Water	In every case, humidity increased but air temperature decreased across the cooling pad because of latent heat addition. Proved the latent heat effects and humidification process.
Gleebpratum and Salakij [86]	EC system with a wetted Salo fabric	Numerical	Thermal resistances network, and MATLAB	Water	Nearly predicted the temperature profile on the rear surface of the PV model.
Abed et al. [87]	EC system with a wetted cotton porous wick	Numerical	Simulation using COMSOL Multiphysics software	Water	Proved that PV panel temperature increases when solar radiation intensifies. An approximately 9.57% reduction in the panel's back side temperature.
Arkar et al. [78]	EC system with a bio-based cotton-cellulose fiber layer	Numerical	Matrix inversion method solved in Microsoft Excel platform	Water	Simulated the temperature of virtual PC cells for different cases. Determined the overheating hours for the virtual PV cells.
Haidar et al. [84]	EC system with co-current configuration of water and air flow	Theoretical	Engineering Equation Solver (EES) software	Water	Illustrated the effects of factors on cooling process. A decrease of about 6 °C in the temperature of the PV panel.
Drabiniok and Neyer [88]	EC using a bionic micro porous evaporative system	Theoretical	Mathematical equations	Water	Proved the benefits of EC on the overall efficiency of the system. A 10% improvement in efficiency for actual roof PV systems when exposed to 100W of incident solar radiation.
Kaneessamkandi et al. [83]	EC system with a heat pipe thermosiphon	Theoretical	Euler's method	Acetone	Provided an analysis of heat balance of the PV panel with EC cooling. A maximum decrease of 10 °C in the PV panel temperature.

amplified the EC impact, leading to a greater equivalent temperature variation across thermoelectric modules. Ultimately, this contributed to an overall improvement in efficiency.

A transient numerical model of the thermal response of vertical PV cell with phase change material was innovated by Arkar et al. [78] to identify the range of properties of phase change material to reveal the efficiency of evaporative cooling technique. The model was created based on the energy balance and a series of finite-difference equations. The equations were solved utilizing the matrix inversion method within a Microsoft Excel platform. The model allowed for simulating the temperature of virtual PV cells in scenarios involving both micro- and macro-encapsulated phase change material. The surface heat transfer coefficients including convective and radiative components on the upper and lower surfaces of the virtual PV cell was established through experimental results, and it were then incorporated into a numerical tool in the shape of multi-parametric approximation models. Also, approximation models for virtual PV cell temperature were developed based on the experimental results. The models were utilized to determine the overheating hours for the virtual PV cells.

Haidar et al. [84] created a theoretical model relying on the principles of heat and mass transfer that take place in the vicinity of the rear surface of a solar PV panel. The model was developed based on the EC operating with co-current configuration of air and water flow. The model included the heat and mass transfer taking place between a water layer and the surrounding air, along with the heat transfer with the PV panel. The acquired results illustrated the impact of certain geometric factors such as air flow rate, temperature, and humidity, on the cooling process. It was observed that there is a notable reduction in the temperature of the PV panel when there was a decrease in the air inlet temperature. Furthermore, there was a considerable rise in air temperature with the increase in solar radiation. As a result, a decrease of approximately 6 °C in the temperature of the PV panel was obtained when the EC was applied.

Drabiniok and Neyer [88] fabricated and tested a theoretical model of bionic evaporation backside cooling. The model was verified by experimental investigation. The theoretical model provided the calculation of the additional benefits of EC on the overall efficiency of the entire system including all pertinent environmental variables. The calculations declared that the impact of EC on the overall cooling balance was highest at low wind speeds due to the limited cooling effect of ventilation. The EC effect was computed by the developed model utilizing the experimental parameters. The calculations were in great agreement with the measurements, stating the theoretical model was suitable to further optimize cooling methods for upcoming highly efficient PV systems. The model indicated a 10% improvement in efficiency for actual roof systems when exposed to 100W of incident solar irradiation while there was a 4.8% efficiency increase in experimental study under low incident solar power of less than 575W.

Kaneessamkandi et al. [83] theoretically and experimentally investigated the effects of heat pipe thermosiphon on the operating temperature of a PV panel. Acetone was used as a coolant. The theoretical investigation included an analysis of the heat balance of the panel with a cylindrical heat pipe in direct contact with the panel's rear surface. The temperature data from experimental results and temperature data from theoretical modeling had a fair agreement for PV panel with EC. The maximum decrease in PV panel temperature was 10 °C with 50 mL acetone.

The summary of numerical and theoretical studies on EC for PV panels is presented in Table 5.

3. General discussion and summary of EC cooling

It was concluded that the EC system was effective to sustain the module temperature in the optimum range. Most of solar irradiance that reaches a PV cell is not converted into electricity, but rather transformed into heat energy, which subsequently raises the temperature of the cell. Elevated temperatures decrease the conversion efficiency and lifetime of PV cells [88–91]. Therefore, cooling is essential to recover the cell's electrical efficiency and prolong its lifetime. EC systems have a huge potential for managing the temperature of PVs operating in hot and arid

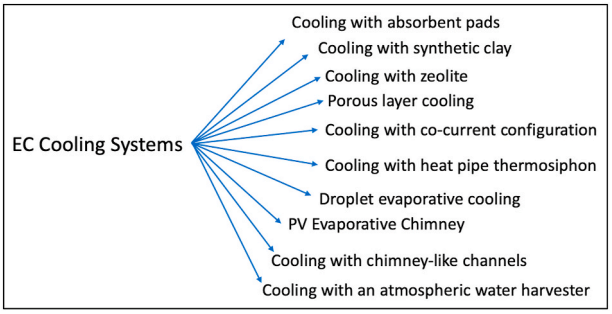


Fig. 22. Existing EC systems.

weather conditions.

Studies have been carried out experimentally, numerically, and theoretically to reveal the effectiveness of utilizing EC methods on PV modules. Some of the studies have investigated the EC for PVs both experimentally and theoretically or numerically to validate results. It was confirmed in experimental studies that the EC systems efficiently lower the operating temperature of PV modules, resulting in improved electricity generation and overall efficiency. Likewise, in the numerical and theoretical studies it was verified that the EC systems reduce the modules' temperature and recover the overall performance and efficiency of PV systems. Moreover, they assert a better understanding of the phenomenon of EC.

Existing EC systems, presented in Fig. 22, use an absorbent pad (wood wool, cellulose pad, cotton wicks, hydrophilic pad, cotton-cellulose pad, burlap cloth) located at rear side of PV modules, synthetic clay, zeolite (a water absorbent material), phase change layer, co-current configuration (created by a fan blowing air inside a duct alongside water flow), droplet evaporation, PV Evaporative Chimney, chimney-like-channels, an atmospheric water harvester and a heat pipe thermosiphon. They mostly utilize water as a coolant and are simply categorized as active or passive according to whether they use an external device such as a pump, fan, in this study. The systems' performances were given in terms of temperature reduction, PV panels' power output and efficiency increase. The variations in solar irradiation were not directly given in some papers; therefore, some of them were obtained from the graphics presented in the studies.

Current EC systems for PV modules mainly use water as a coolant because it is relatively cheap, easy to access, environmentally friendly, versatile, does not undergo any chemical reaction with the PV module, non-toxic, and has a high heat capacity. Nanofluids (Copper Oxide/water, and Aluminum Oxide/water), ethanol and acetone have been used as alternative to water.

General observations and trends found from the experimental application can be summarized below;

- Temperature Reduction: Many of the systems, both passive and active, demonstrate substantial reductions in temperature, ranging from 8 °C to as high as 40 °C. This is a crucial factor for enhancing the performance and lifespan of PV modules.
- Efficiency Improvement: Across various systems, improvements in electrical efficiency and power output are consistently observed. These enhancements contribute to better overall energy harvesting from the solar panels.
- Coolant Diversity: Water remains a popular coolant, known for its accessibility and heat absorption properties. However, there is a trend towards exploring alternative coolants such as ethanol, acetone, and zeolite, showcasing a pursuit of efficiency and environmental considerations.
- Material Innovation: The use of innovative materials, including synthetic clay, cotton wicks, cloth, rubber pipes, zeolite, atmospheric water harvester, cotton-cellulose fiber, and cellulose pads, highlights the importance of material used in designing effective EC systems.
- Passive vs. Active: Both passive and active systems show effectiveness in reducing temperatures and improving efficiency. Passive systems often leverage natural processes, making them simpler and potentially more sustainable. Active systems, on the other hand, provide more dynamic control and can be optimized for specific conditions.
- Hybrid Approaches: The hybrid system (PV/EC) and the use of both passive and active elements in some systems suggest a trend towards combining the strengths of different approaches for optimized performance.

Table 6
Performance improvement, advantages, disadvantages, implementation cost, and limitations of the reviewed approaches of EC for PV module cooling.

Aspect	Summary	Key Points
Performance Improvement	Demonstrated temperature reduction enhances electricity generation and overall efficiency of PV modules.	- Significant reduction in operating temperatures. - Consistent improvement in performance observed in experimental studies.
Advantages	Versatile system configurations, cost-effective primary coolant (water), and flexibility with alternative coolants contribute to favorable attributes.	- Different options with absorbent pads, synthetic clay, zeolite, etc. - Water: cost-effective, environmentally friendly, and non-reactive. - Alternative coolants (nanofluids, ethanol, acetone) offer flexibility. - Avoidance of toxic chemicals and expensive materials for cost-effectiveness.
Disadvantages	Challenges in water-scarce regions and potential environmental concerns due to increased water consumption. Variations in solar irradiation impact consistency across locations.	- Water scarcity challenges in some regions. - Environmental concerns related to increased water consumption. - Variations in solar irradiation may affect performance consistency.
Implementation Cost	Generally favorable due to the use of common materials, avoidance of toxic chemicals, and cost-effective coolants. Integration of innovative technologies may introduce additional costs.	- Cost-effectiveness from common materials and water use. - Avoidance of toxic chemicals supports affordability. - Integration of innovative technologies may introduce additional costs.
Limitations	Environmental factors affecting performance, optimization challenges for various PV technologies, and water dependency pose constraints.	- Performance influenced by environmental factors. - Challenges in optimizing configurations for different contexts. - Water dependency poses constraints in water-scarce regions.

Sustainability: Several systems incorporate sustainable and bio-based materials, aligning with the growing emphasis on environmentally friendly technologies. This includes the use of cotton, cellulose, and bio-based fibers.

In conclusion, the different range of EC systems presented in the studies reflects the ongoing innovation in PV panel EC technologies. While each system has its unique features, the overarching goal is to improve the efficiency and longevity of PV panels under varying environmental conditions. The choice between passive and active systems, as well as the selection of coolant and materials, depends on factors such as cost, accessibility, and specific application requirements. The

Table 7
Different chemicals with their boiling points and safety and toxicity features.

Chemical	Boiling point (°C)	Safety and toxicity
Diethyl ether $(C_2H_5)_2O$	34.6	Highly flammable, harmful to skin
Dichloromethane (CH_2Cl_2)	39.6	Serious health risks, carcinogenic, absorbed through skin Has a relatively low flash point
Carbon disulfide (CS_2)	46.26	Flammable, Eye and skin irritation
Acetone $((CH_3)_2CO)$	56.1	Highly flammable. Eye and skin irritation
Ethanol (C_2H_6O)	78.37	Highly flammable, no significant health risks
Isopropyl alcohol $(CH_3)_2CHOH)$	82.6	Flammable Eye and skin irritation.

continuous exploration of novel materials and configurations indicates a commitment to advancing the field of solar energy and promoting sustainable practices.

The current EC systems do not utilize any toxic chemicals, very expensive materials, and components. Table 6 provides a concise overview of the key points related to performance improvement, advantages, disadvantages, implementation cost, and limitations of the reviewed approaches for PV module cooling.

4. Recommendations

Necessary modifications can be made in an EC system to increase overall efficiency. Based on the existing EC systems for PV modules, the recommendations are outlined as followed.

- Most of the EC systems for PVs are left open and exposed to the surrounding air; therefore, there is not a re-circulation of the coolant, causing water waste. Water vapor could be collected by condensation into a storage tank, and condensed water can be recirculated in the cooling systems. Condensation of water will allow the cooling systems to be used in areas where water scarcity is prevalent. For example, an underground cooling system can be designed to condense the water vapor without utilizing any energy generated by the PVs.
- An EC system can be integrated into PV panel production. For instance, Drabiniok and Neyer [92] innovated and fabricated a bionic micro porous evaporative polymer which produces a significant uniform cooling effect by the evaporation of water through micro pores in the polymer foil. Results of the study claimed that the polymer foil could be integrated into the production of PVs. In addition, the porous layer invented by Wang et al. [67] could be incorporated into the manufacturing process of PV modules. Moreover, Wang et al. [93] introduced a metal foam porous-wall mini--channel heat sink for PV systems with high concentration ratio and strong heat generation. R141b was used as working fluid and its boiling characteristics under the electric field were investigated. They proved that the mini-channel heat sink can enhance the energy conversion efficiency of the linear Fresnel CPV system. Therefore, this system could be integrated into fabrication of CPV panels, as well.
- Alternative coolants having low evaporative temperature and high heat capacity can be used in experiments and compared with water in future studies. Various chemicals with their evaporation temperatures at standard atmospheric pressure, as well as safety and toxic properties are presented in Table 7; created based on information found on National Library of Medicine website [94]. Ethanol and isopropyl alcohol could be used in EC systems for CPV systems because they are safe and non-toxic compared to other chemicals. Moreover, due to having a low flash point, dichloromethane, diethyl ether, carbon disulfide could be good alternatives for PV modules if proper precautions and safety measures are taken. Acetone has recently been employed as coolant in an EC system that maintained the PV module temperature in the ideal range. Also, ethanol has already been used in the phase change cooling system that significantly improved the electrical power output and efficiency of the PV module.

The subjects presented in Table 8 summarizes the key recommendations for enhancing the efficiency of existing EC systems for PV modules.

5. Conclusion

In the literature, there is a sufficient amount of EC systems utilizing different materials, techniques, and configurations for PV modules.

The EC systems mentioned in this work was experimentally, numerically, and theoretically evaluated in the study. The following statements delineate the concluding remarks.

- The EC have a great potential in enhancing the overall performance of PV systems. They effectively reduce the operating temperature of

Table 8
Summary of recommendations and their explanations.

Recommendation	Descriptions	Benefits	Considerations
1. Condensation and Recirculation System	Implement condensation to collect water vapor. Recirculate condensed water in cooling systems. Design underground systems for water condensation.	Water conservation, suitable for water-scarce areas. Can operate without PV-generated energy.	Assess feasibility and maintenance requirements of underground systems.
2. Integration of Evaporative Material	Integrate a suitable evaporative material into PV panel production. Use heat sinks for efficient cooling.	Uniform cooling effect, potential efficiency gains. Diversify cooling technologies.	Evaluate manufacturing costs and scalability of integrated technologies.
3. Experimentation with Alternative Coolants	Experiment with coolants having low evaporative temperature. Compare performance with water in PV module cooling. Prioritize safe options like ethanol and isopropyl alcohol.	Potential efficiency improvements. Safety and non-toxicity. Exercise caution with flammable and harmful chemicals.	Consider environmental impact and regulations along with economic impact in coolant selection.
4. Assessment of Chemical Options	Evaluate chemicals based on boiling points, safety, and toxicity. Prioritize safe and non-toxic options. Perform risk assessments before implementation.	Informed selection of alternative coolants. Minimize health and environmental risks. Thoroughly assess safety measures.	Consider environmental impact and regulations along with economic impact in coolant selection.

PV modules, resulting in improved electricity production and increased efficiency.

- Current EC methods utilize absorbent pads, synthetic clay, zeolite, phase change layers, co-current configurations, droplet evaporation, PV Evaporative Chimney, chimney-like channels, an atmospheric harvester, and a heat pipe thermosiphon.
- To numerically study the thermal behavior of evaporatively cooled PV modules and introduce the possible benefits of EC systems for PV modules, thermal, multi-parameter, simulation models have been developed.
- To approve the experimental scenarios and present the factors effecting the EC of PV systems, theoretical investigations have been carried out.
- Existing EC systems for PV modules mostly use water as a coolant.
- The EC systems can be passive or active based on whether they use an external device.
- EC designs using absorbent pads seems to demonstrate promising results in temperature reduction and efficiency increase. Particularly, EC system utilizing cotton wicks immersed into pure water displayed the best outcomes in terms of PV modules' output and efficiency.
- The EC systems can be incorporated into the manufacturing process of PV modules.
- Ethanol and isopropyl alcohol could be used in EC systems for CPV modules because they are relatively safe and non-toxic.
- Because of having low boiling point, dichloromethane, diethyl ether, and carbon disulfide could be good alternatives to water as a coolant for PV systems if proper precautions and safety measures are taken.
- Acetone was recently employed as coolant in an EC system and maintained the PV module temperature in the ideal range.
- Ethanol has already been used in the phase change cooling system that significantly improved the electrical power output and efficiency of the PV module.
- The current EC systems for PVs appear to be environmentally friendly, inexpensive, and reliable.
- The number of studies on EC systems for PVs have a trend of increasing.

Declaration of competing interest

There is no conflict of interest.

Data availability

No data was used for the research described in the article.

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